Beyond Open Models: Building Open Web Search Infrastructure for AI Agents

Michael Granitzer
Chair of Data Science, University of Passau, Germany
IT:U Interdisciplinary Transformation University Linz, Austria
Coordinator OpenWebSearch.EU

CLEF 2026 · Keynote · 22 September 2026

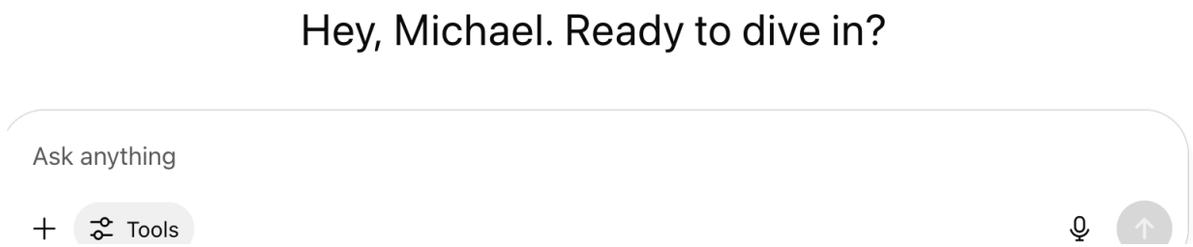
Funded by

Bundesministerium für Forschung, Technologie und Raumfahrt

Bundesministerium für Landwirtschaft, Ernährung und Heimat

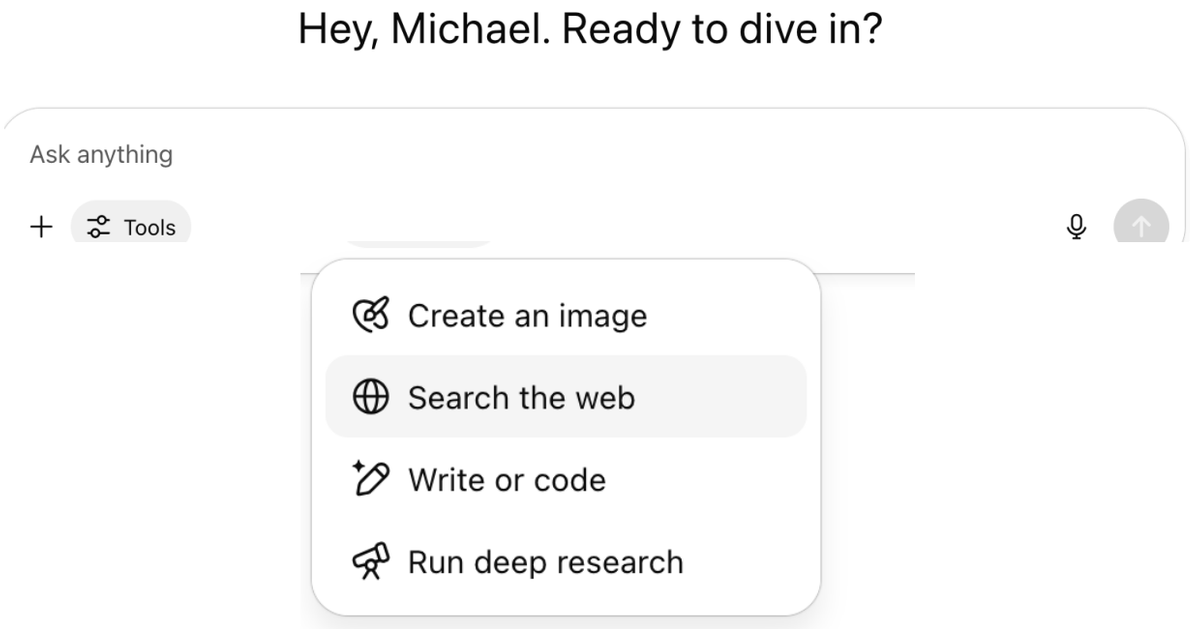
Bayerisches Staatsministerium für Wirtschaft, Landesentwicklung und Energie

Search is dead



Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

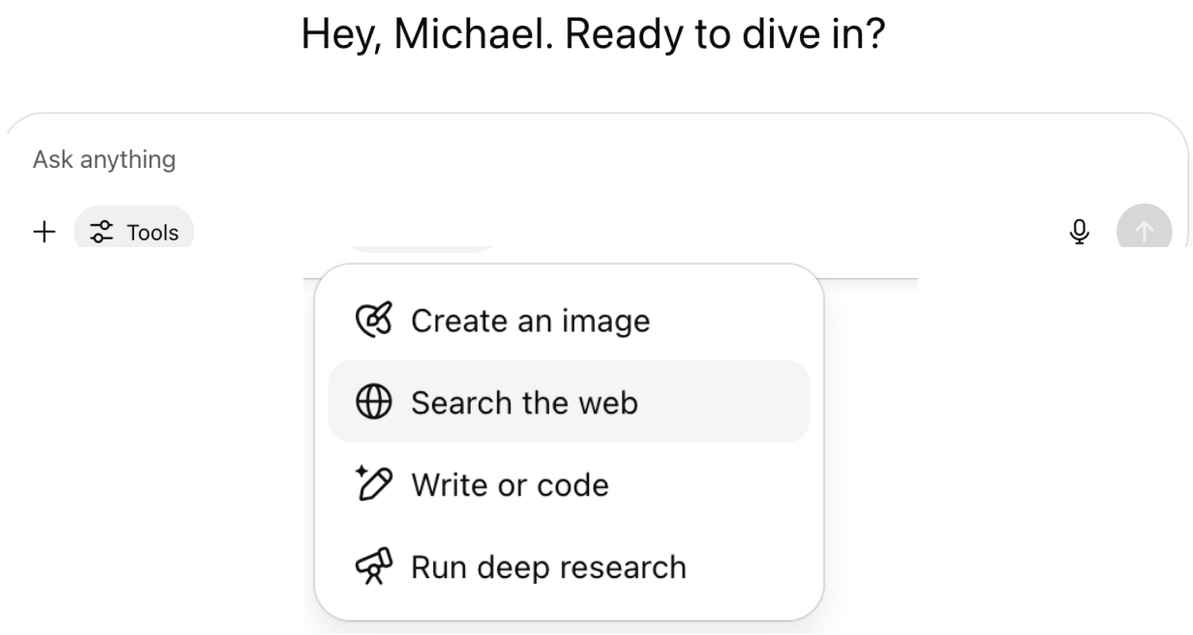
Search is dead



"Search the web" sits behind the box, switched on by default in most assistants.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search



"Search the web" sits behind the box, switched on by default in most assistants.

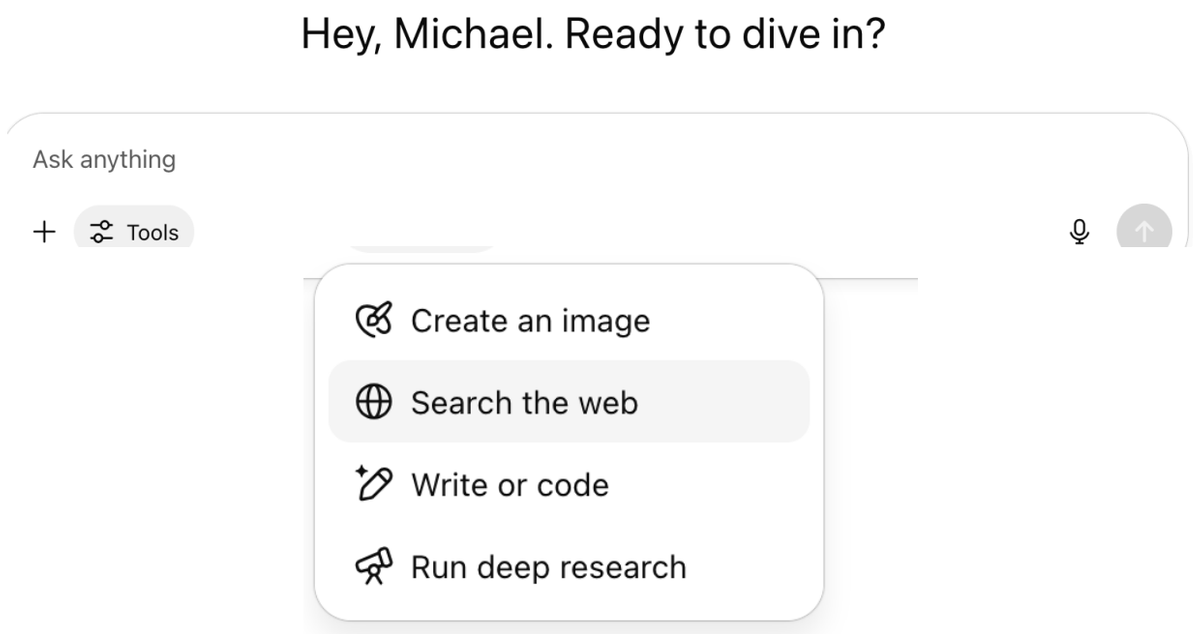
34.5%
of ChatGPT queries switched on web search in February 2026

14% → 24%
of ChatGPT usage is "seeking information" in one year, "a very close substitute for web search"

20–26
search calls per question by a deep-research agent

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search



"Search the web" sits behind the box, switched on by default in most assistants.

34.5%
of ChatGPT queries switched on web search in February 2026

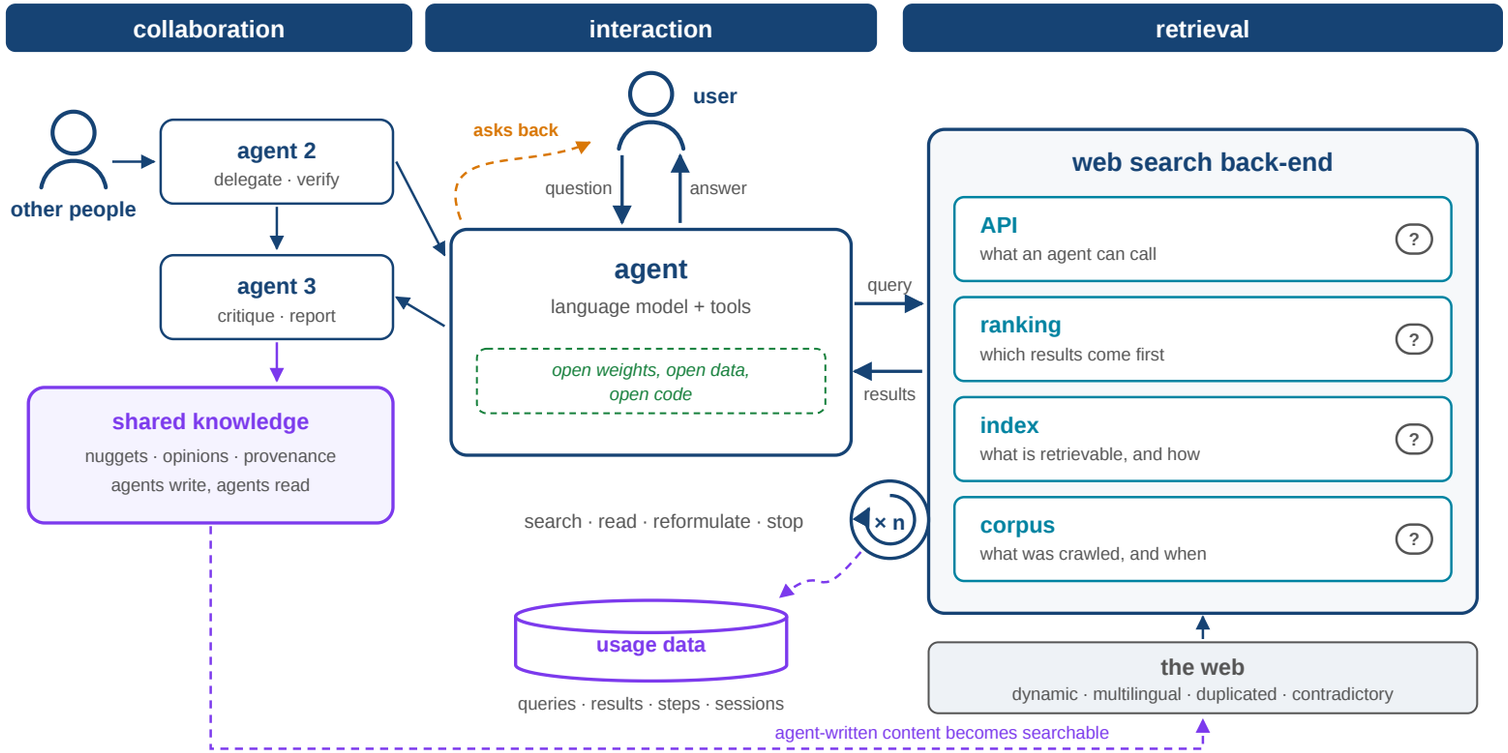
14% → 24%
of ChatGPT usage is "seeking information" in one year, "a very close substitute for web search"

20–26
search calls per question by a deep-research agent

The box where people type changed. What answers them is still a search engine, queried more often, by something that is not a person.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Agentic (search) (eco)systems are complex



> And this is only the search part. Beyond it the system grows again: tools and MCP, skills and skill learning, workflows and orchestration.

Understanding search meant operating search

How did we come to understand web search users, and the needs of web search systems?

> We did learn mostly from query logs and obtaining them was hard. But a search session is richer than the query and the click it leaves behind.
>
> The richest understanding came from operating a system itself, which allows to vary all variables (e.g. A/B-testing).

A released web log records two kinds of event

AOL 2006, RELEASED AND WITHDRAWN IN THREE DAYS

A query, and a click on a result. 36.4M queries, 657,426 users, 19.4M clicks — and the only results in the file are the ones that were clicked. Sessions are not marked.

A system you run records fifty-eight

SOWIPORT, OPERATED BY GESIS, 2014–15

Typing a query, selecting a facet, visiting a document, following a citation, exporting a record. One year gave **558,008 sessions** and about **8M interactions**, of which 179,796 were queries — and the result lists are stored as they were shown.

Reimer et al., The Archive Query Log, SIGIR 2023, Table 1 (AOL and ORCAS figures) · Mayr, Sowiport User Search Sessions Data Set, 2016, doi:10.7802/1380 · Joachims, KDD 2002 · Joachims, Swaminathan, Schnabel, WSDM 2017.

One hypothesis and four topics for today

HYPOTHESIS

We have to build usable agentic search systems at scale in order to understand them fully.

- Components can be studied on static collections, in isolation.
- Systems cannot: what they do depends on the corpus, code, people and agents using them.

TOPIC 1

Building a sovereign search system

OpenWebSearch.EU and the Open Web Index: web data, an index and a search API that a research community owns and operates.

TOPIC 2

Corpora for evaluation, and scale as a variable

The collections an open index lets a lab cut for itself, and what they make measurable once corpus size and text quality can be varied.

TOPIC 3

Usage data and simulation

We missed much of the interaction data of web search. We should not miss the interaction data of real users with agentic search systems.

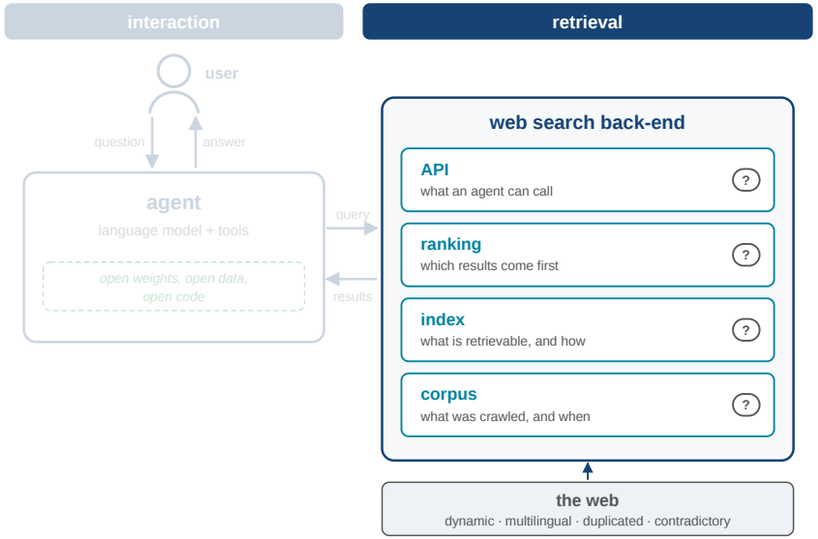
TOPIC 4

Societies of humans and models

Beyond one user and one system: many agents, many people, and the knowledge they produce between them.

Topic 1 · Building a sovereign search system

OPENWEBSEARCH.EU, THE OPEN WEB INDEX, AND A SEARCH API ON TOP



Looking at the Model-side first

Open weights

Llama, Qwen, DeepSeek, Mistral and gpt-oss can be run and fine-tuned by anyone. Their training data and recipes stay private, so they can be used but not inspected.

Fully open

OLMo 3, Apertus, Soofi S, OpenEuroLLM Pleias, Marin and Comma release weights, data, code and checkpoints.

Open-source models on top of closed retrieval infrastructure remain closed research instruments.

OSI, Open Source AI Definition 1.0, Oct 2024 · AI2, OLMo 3, Nov 2025 · Swiss AI, Apertus, Sep 2025 · SOOFI, Soofi S, Jul 2026.

Today's API-based back-ends are not open

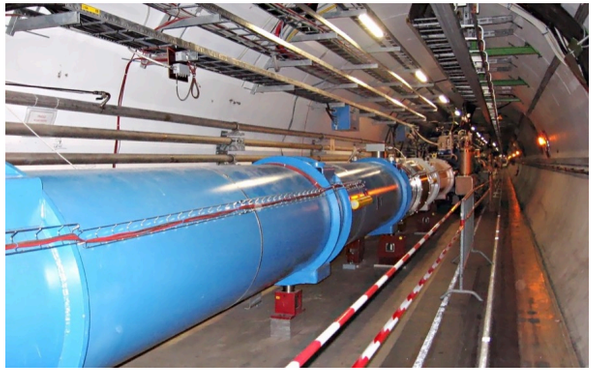
Service	Still available	Corpus known	Ranking replaceable	Snapshot fixable	Logs for research
Google Custom Search API	ends Jan 2027	✗	✗	✗	✗
Bing Web Search	retired Aug 2025	✗	✗	✗	✗
Brave Search API	✓	claims 40B pages	re-rank only	✗	✗
Exa · Tavily · Perplexity	✓	✗	✗	✗	✗
OpenAI · Anthropic search tools	✓	✗	✗	✗	✗

> So we do not have ability to inspect, reproduce, vary, operate and evaluate the whole information-access process, which means we lack **research sovereignty**.

Research sovereignty

Photographs: J. Herzog CC BY-SA 3.0 · ITER Organization CC BY 2.0 · NASA/MSFC/D. Higginbotham · Rama CC BY-SA 2.0 fr. Wordmarks belong to their owners.

Research sovereignty



CERN · particle physics.
25 member states, one accelerator.



ITER · fusion.
35 nations, one reactor none could build alone.



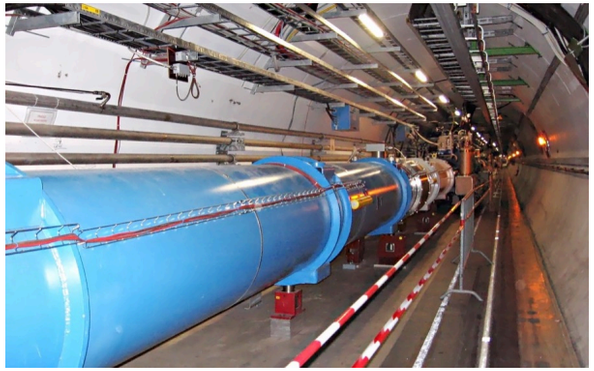
Webb · astrophysics.
Three agencies fly it, the observations go to a public archive.



Copernicus · climate.
Earth observation, free and open by policy.

Photographs: J. Herzog CC BY-SA 3.0 · ITER Organization CC BY 2.0 · NASA/MSFC/D. Higginbotham · Rama CC BY-SA 2.0 fr. Wordmarks belong to their owners.

Research sovereignty



CERN · particle physics.
25 member states, one accelerator.



ITER · fusion.
35 nations, one reactor none could build alone.



Webb · astrophysics.
Three agencies fly it, the observations go to a public archive.



Copernicus · climate.
Earth observation, free and open by policy.

Google
the web index, and the ranking

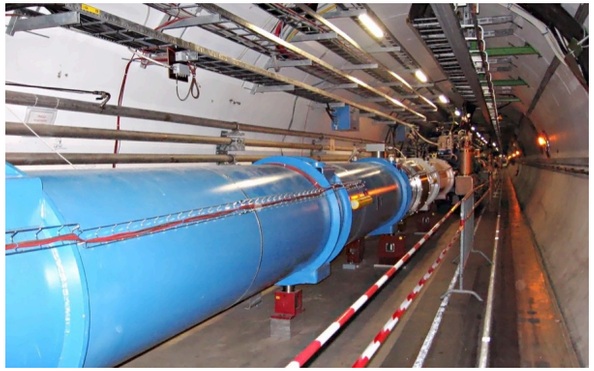
Meta
the social record

Hugging Face
the models and the datasets

the code

Photographs: J. Herzog CC BY-SA 3.0 · ITER Organization CC BY 2.0 · NASA/MSFC/D. Higginbotham · Rama CC BY-SA 2.0 fr. Wordmarks belong to their owners.

Research sovereignty



CERN · particle physics.
25 member states, one accelerator.



ITER · fusion.
35 nations, one reactor none could build alone.



Webb · astrophysics.
Three agencies fly it, the observations go to a public archive.



Copernicus · climate.
Earth observation, free and open by policy.

Google
the web index, and the ranking

Meta
the social record

Hugging Face
the models and the datasets

the code

> **Our field's instruments are privately owned. No control, no costs.** What could go wrong?

Photographs: J. Herzog CC BY-SA 3.0 · ITER Organization CC BY 2.0 · NASA/MSFC/D. Higginbotham · Rama CC BY-SA 2.0 fr. Wordmarks belong to their owners.

What could go wrong

Every one of these was, at the time, infrastructure that somebody's research depended on.

2018
Microsoft buys GitHub for **7.5 bn USD**

2023
Twitter ends free API access; Reddit prices its own

2023
GPT-4: no architecture, data or training method disclosed

2024
Meta shuts down CrowdTangle

2025
Bing Search APIs retired

2026
NVIDIA agrees to acquire Hugging Face for **12.9 bn USD**

> Openness of an artefact is not enough.
>
> Research sovereignty requires a jointly managed public infrastructure.

Microsoft newsroom, 4 Jun 2018 · Murtfeldt et al., RIP Twitter API, 2024 · OpenAI, GPT-4 Technical Report, arXiv:2303.08774 · Microsoft lifecycle notice, May 2025 · NVIDIA blog and Form 8-K, 2–3 Sep 2026 · Zenodo, operated by CERN with OpenAIRE and EU funding.

OpenWebSearch.EU: building the index together



Objectives

1. Build an open index of the Web
2. Build a corresponding web search infrastructure
3. Establish a network of infrastructure partners.
4. Bootstrap an ecosystem

Sep 2022
project start, Horizon Europe

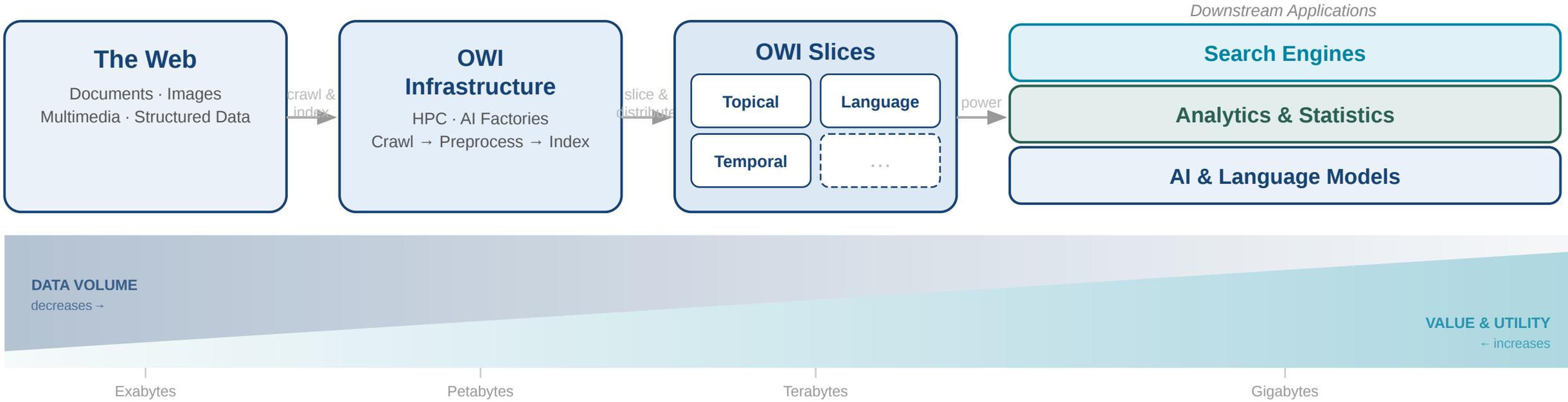
2023
first crawlers and pipelines on the HPC centres

2024
daily index products, dashboard, CLI

2025
datasets, search prototypes, tooling

Feb 2026
project end; operation continues

The Open Web Index: web data every day, ready to use



1,507 TB
crawled

57.6 TB
index

1,901
public datasets

20–100 TB
crawled per month

CIFF
A sparse inverted index, tokenizer included.

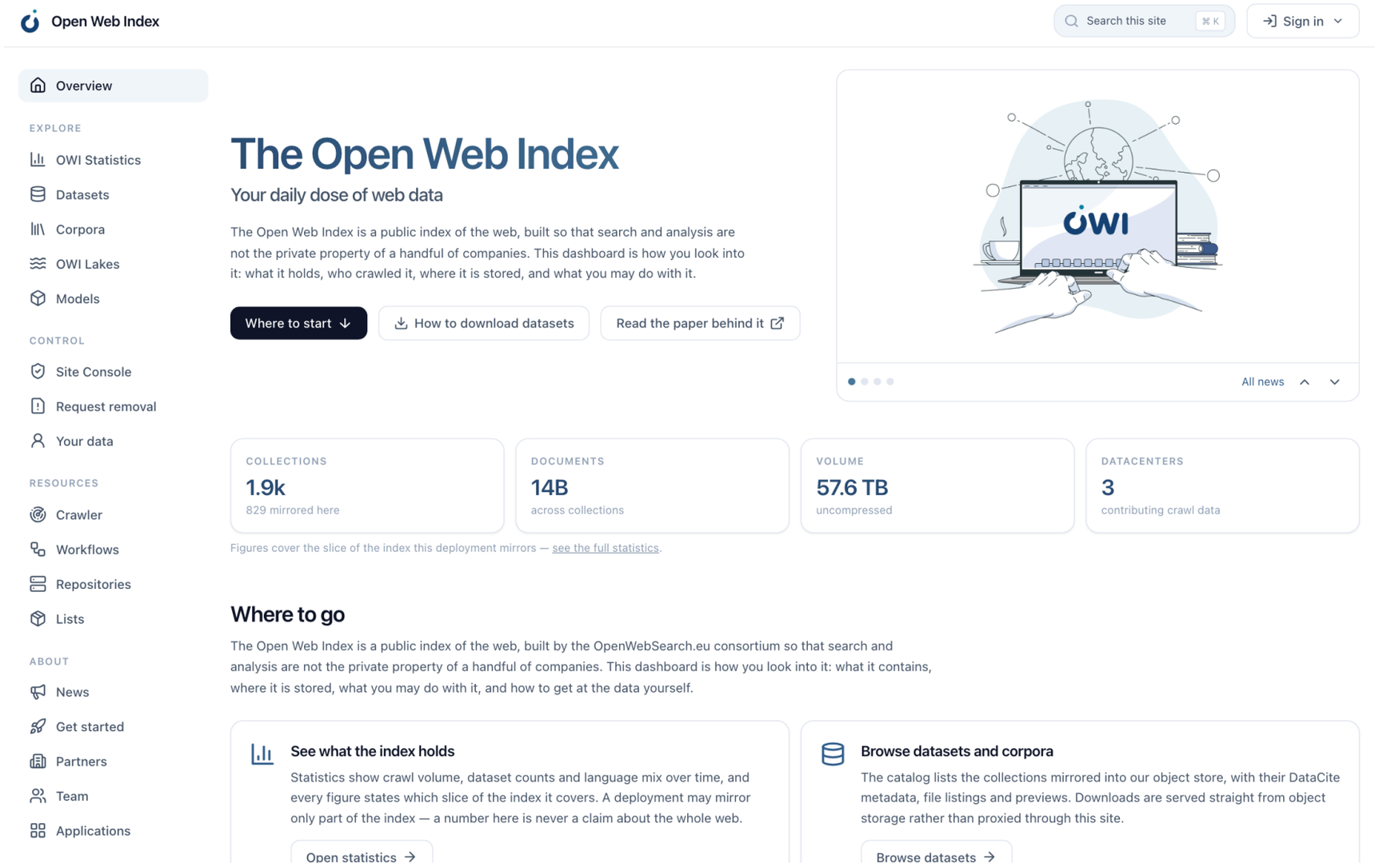
Parquet
Text, language, topic, structured data, locations, technical metadata

Embeddings
Dense Jina vectors on selected subsets.

Federated Infrastructure
3 HPC centres / 2 AI Factories in a extensible federation - hard technical challenge.

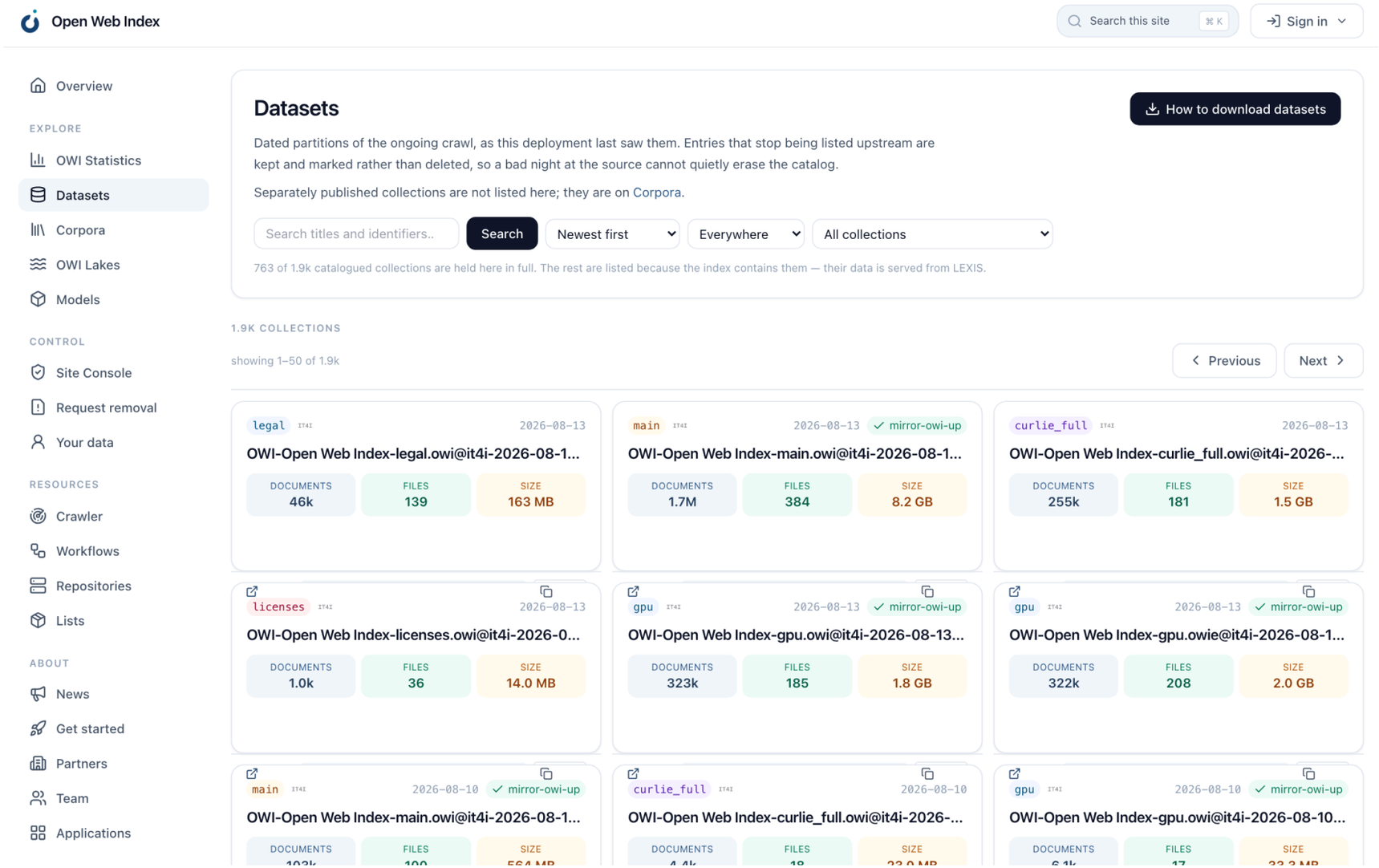
Dashboard figures May 2026 (openwebindex.eu).

OpenWebIndex.eu: Dashboard and Access



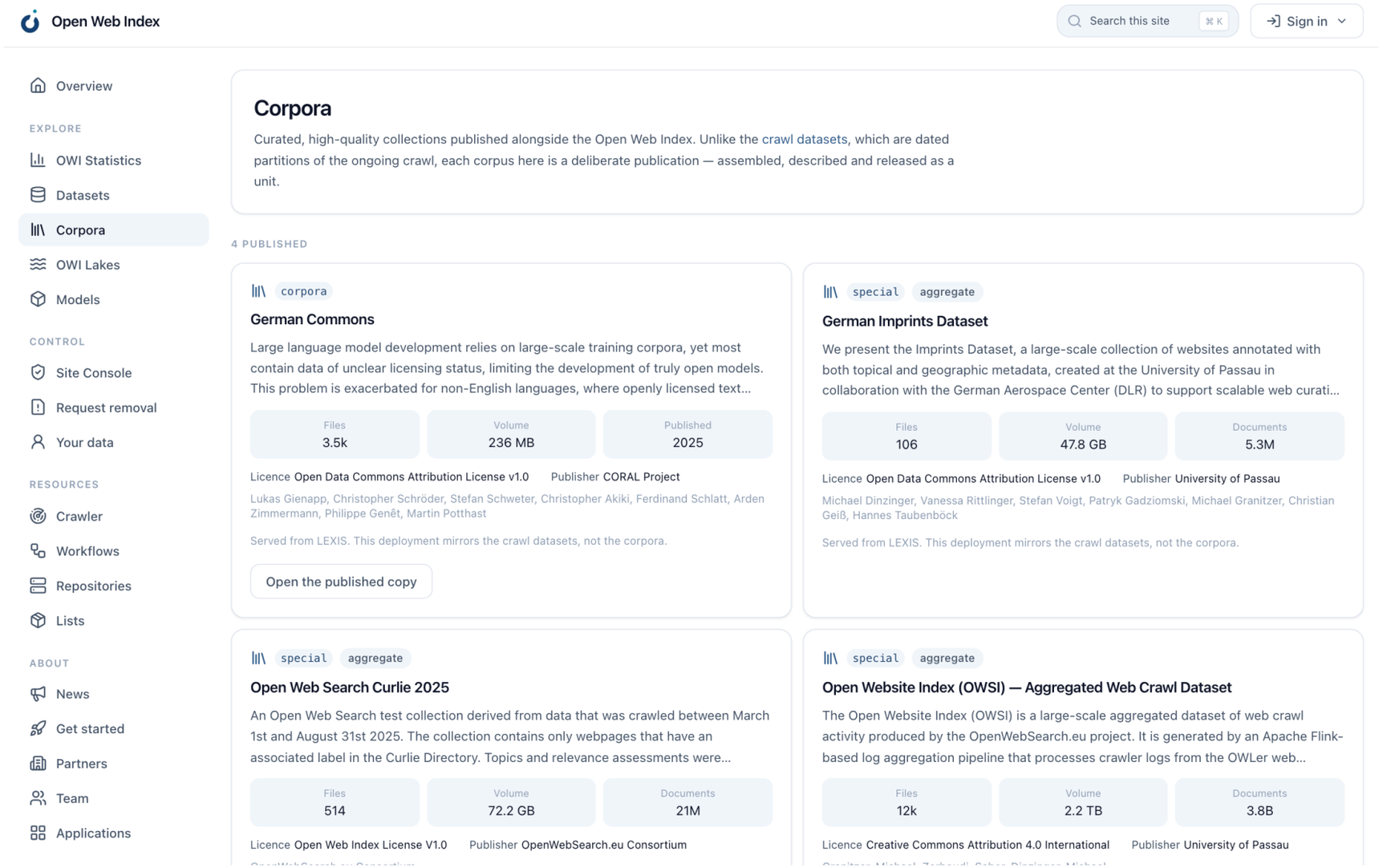
openwebindex.eu

OpenWebIndex.eu: Dashboard and Access



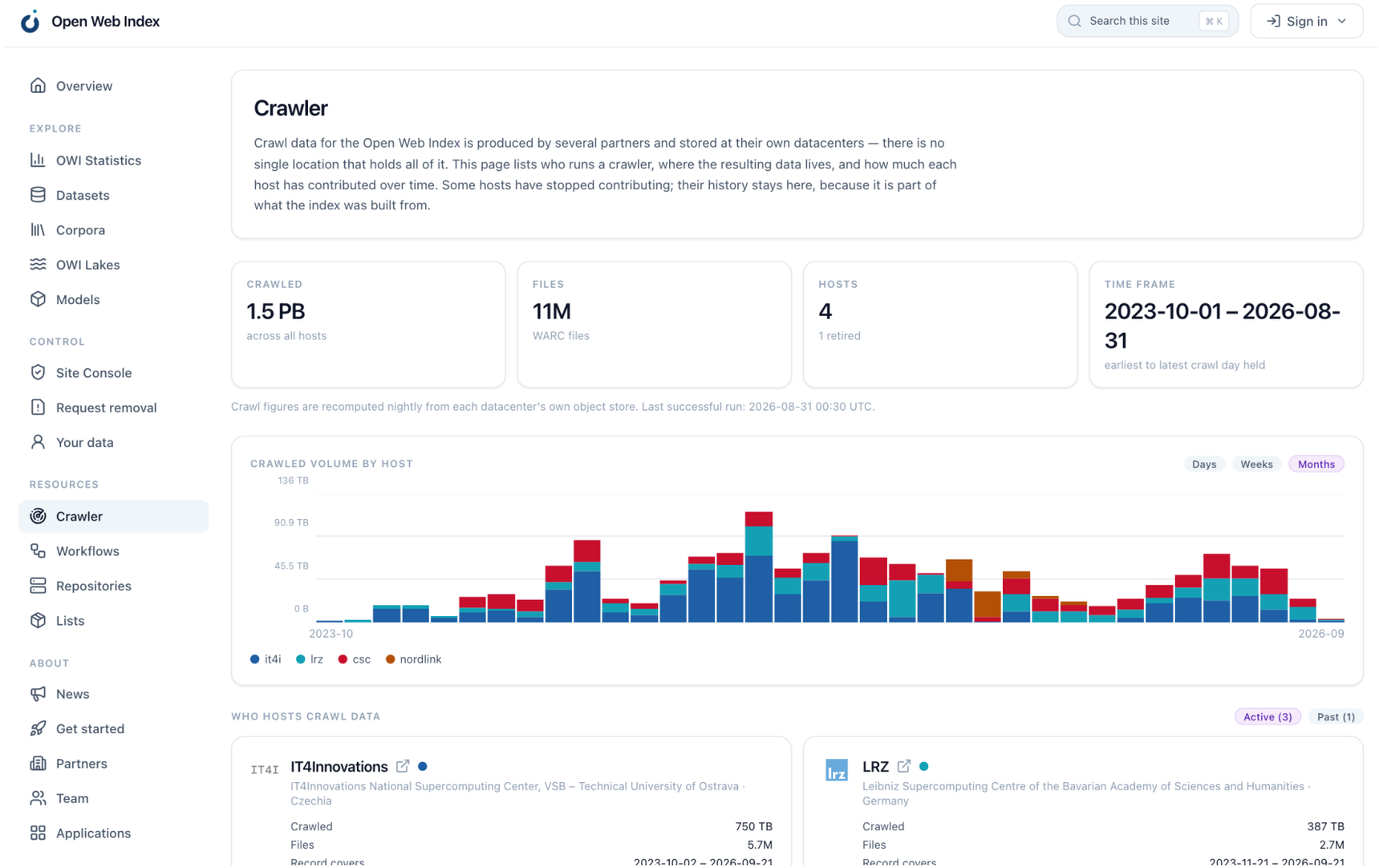
openwebindex.eu

OpenWebIndex.eu: Dashboard and Access



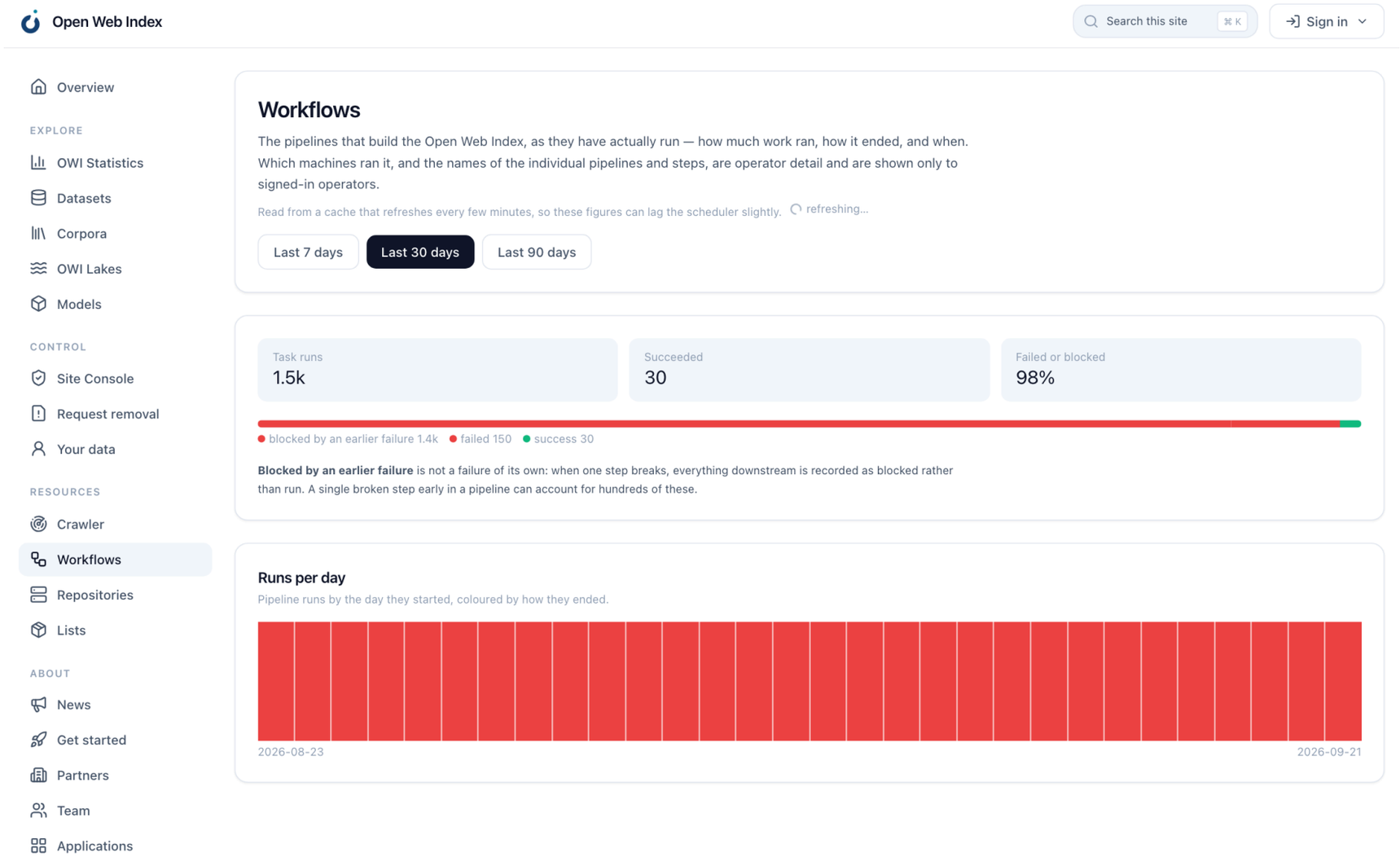
openwebindex.eu

OpenWebIndex.eu: Dashboard and Access



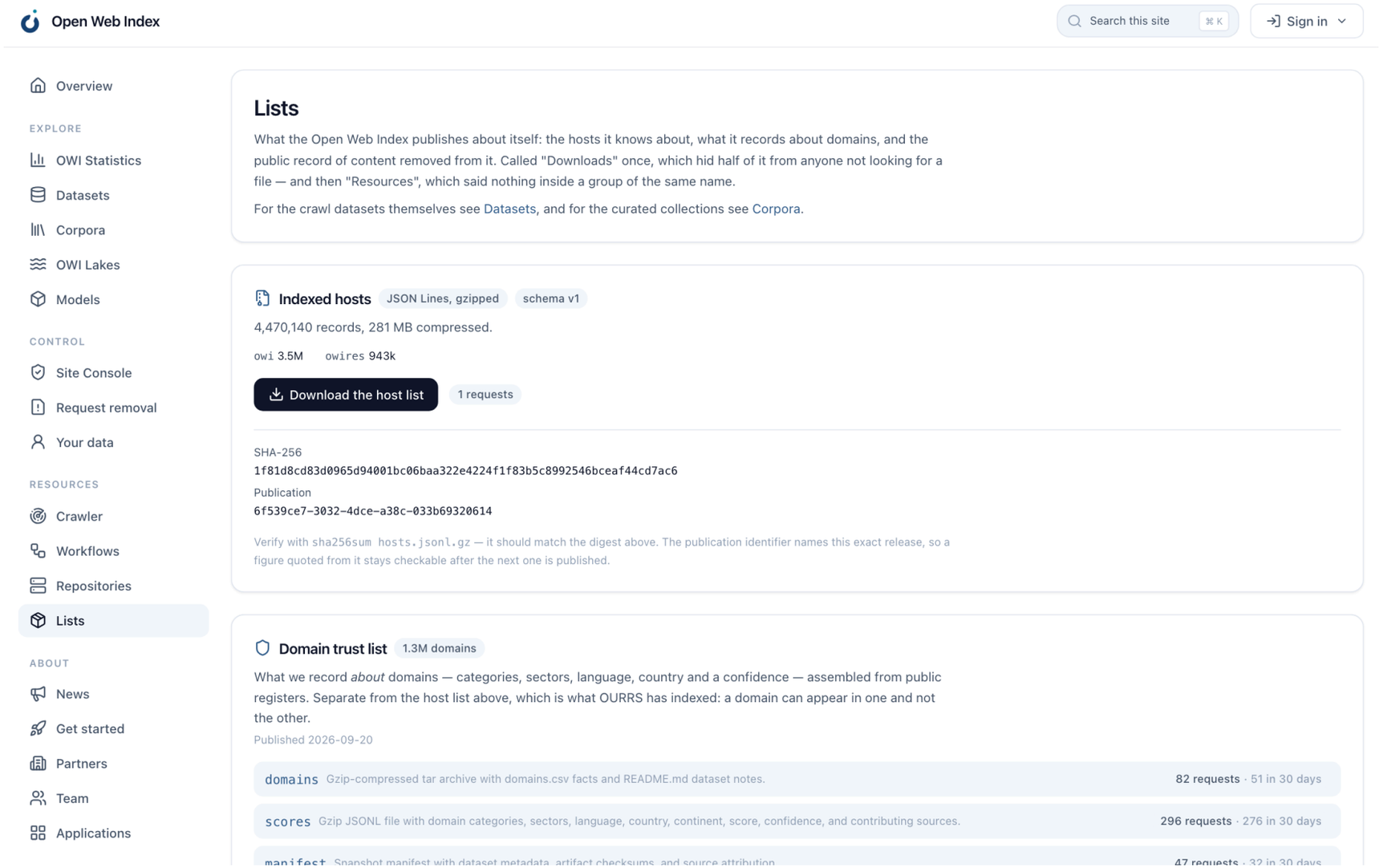
openwebindex.eu

OpenWebIndex.eu: Dashboard and Access



openwebindex.eu

OpenWebIndex.eu: Dashboard and Access



openwebindex.eu

Two ways in: pull the index, or query it where it lies

owilix: the client for the Open Web Index

git-style pull · build · push | slice by language, Curlie topic, domain or site list

1 · pull the partition

your machine or cluster

- year=2025/month=09/day=17/language=de
- year=2025/month=09/day=18/language=de
- year=2025/month=09/day=18/language=en

Partitioned by date and language, so the slice an experiment needs is a path. Partitions merge by union.

2 · query it where it lies

```
s3://openwebindex/owic25/
├ postings/*.parquet     term · doc · impact          always read
├ dict/*.parquet         term → id, df                dropped by NoDict
└ docs/*.parquet         url, length, metadata        dropped by NoDocs
```

DuckDB, on your side · BM25 written as SQL

reads only the postings a query touches; the ranking is yours to change

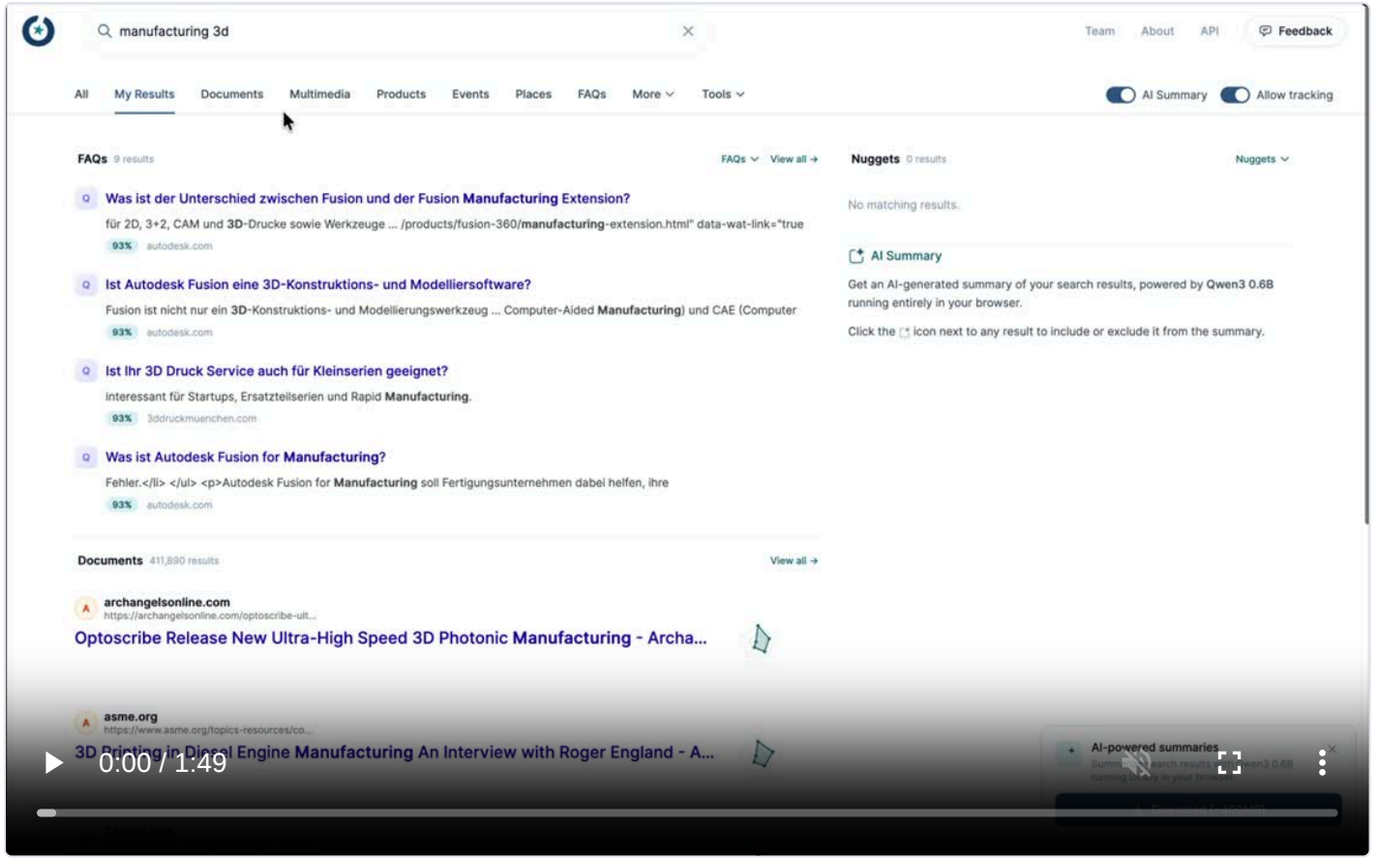
NoDocsNoDict reads the postings and nothing else: most queries 5–10 s, a full topic set in 10–15 min, 45 s with the storage 650 km away.

ECIR 2026 · HENDRIKSEN, HIEMSTRA, DE VRIES

Open Web Indexes for Remote Querying

QUESTION	Can a web-scale index be shared as plain files and queried where it lies, instead of behind an API?
SETUP	ClueWeb09-B, 12-B and 22-B as compressed Parquet on S3; BM25 run client-side in DuckDB; four schemas, each dropping one join.
CONCLUSION	Seconds, not milliseconds. Nobody has to host a copy, and the retrieval model, the snapshot and the sub-collection are yours.

OURRS.eu: the open web search API plus MCP



ourrs.eu, index snapshot September 2026; screencast September 2026.

1 Billion
pages

950M
typed entities

15
verticals, plus an API and MCP

4
isolated corpora

Persisted AI Summaries

Agentic Relevance Feedback

> The Open Web Index Research Retrieval System - OURRS: an early web search prototype over a billion pages of the index (September 2026)

Where the OWI goes next

Scaling up and out

- **Quantity** — crawling across more European data centres
- **Quality** — headless rendering, better crawling strategy
- **Variety** — multimedia and multimodal data
- **Tooling** — better CLI, possible access to raw data, corpus and model management
- **Cross-cutting** — identity, access management, community contributions

OWI Lakes

- Topically coherent, deduplicated, structured sub-indices
- Every index a Parquet data lake on object storage, queried with SQL: a remote representation of a prebuilt index rather than a copy to host
- Community-driven, with the OWI as the backbone

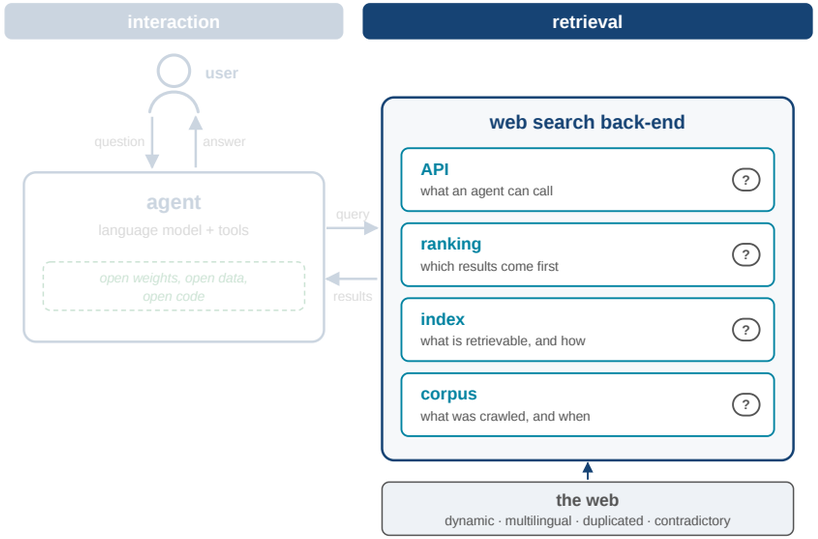
And other verticals

starting with misinformation and disinformation detection.

Hendriksen, Hiemstra, Lin, de Vries, Geese Belong in (Data) Lakes: An Index Lake for Anserini, CIKM 2026.

Topic 2 · Corpora for evaluation, and scale as a variable

WHAT AN OPEN INDEX LETS A LAB CUT FOR ITSELF, AND WHAT THAT MAKES MEASURABLE

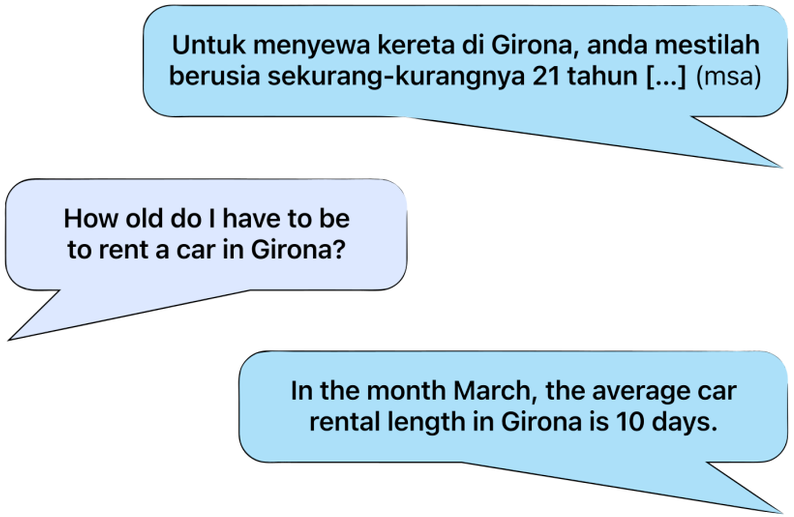


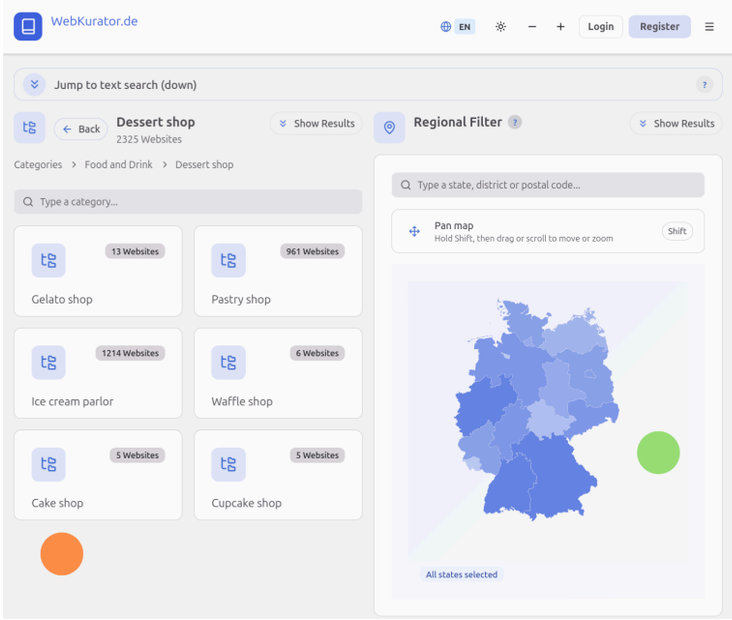
Today's corpora, and what an open index adds

Collection	Documents	Crawled	What the text is
MS MARCO passage	8.8M passages	2016–2018	cleaned passages
ClueWeb22	10B pages, 207 languages	once, in 2022	raw pages with DOM annotations, under licence
MS MARCO Web Search	a 100M-document benchmark on ClueWeb22	once, in 2022	the same pages, with click labels
LongEval web	dated snapshots, 2022–2023	from a search-engine partner	a partner's documents
BrowseComp-Plus	100,195 documents	fixed, 2025	curated so agents can be compared
Open Web Index	**39B URLs crawled, 14B indexed, 57.6 TiB index**	**every day, 3+ more years**	**crawled pages, 185 languages, structured data**

The OWI adds freshness, because crawl and index run daily; scale, because it is web-scale and growing; richness, because structured data comes with it. It is a community effort, so contributions and requests are welcome.

Two example datasets from the index





SIGIR 2026 RESOURCE

WebFAQ 2.0 · multilinguality

BUILT FROM schema.org FAQPage markup, mined and crawled directly. The pairs are written by site owners, not generated.

SIZE 198M question–answer pairs in 108 languages, 14.3M of them aligned across 3,970 language combinations.

CONTROLS which language, or which language pair, an experiment runs in. 1.25M queries ship with 200 hard negatives each.

WEBSCI 2026

WebKurator · region and topic

BUILT FROM the legal imprint of 5.54M German-language sites, geocoded and then topic-labelled automatically.

SIZE 2.58M sites in Germany carry both a location and a topic at launch.

CONTROLS region and topic as two independent axes, so "this topic in that region" is a query, not a filter over one hierarchy.

OWS-Curlie-2025: the collection behind WOWS 2026

The collection

- Six months of our own crawl, March to August 2025, English pages of Curlie-listed sites.
- **20.7M documents**, about 70 GB, with embeddings for the judged parts.
- Three judged splits of 60 to 90k documents. One ships its qrels; two keep them withheld.
- `owilix` to pull it, `ir-datasets-owi` to use it.

The shared task

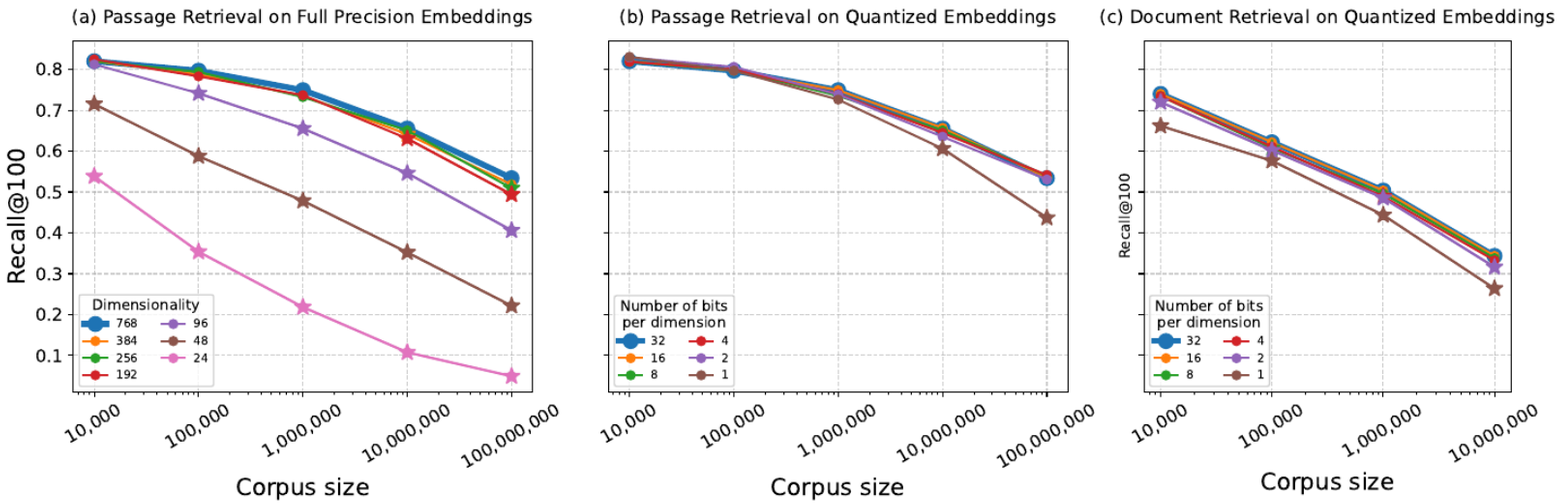
- **WOWS 2026**, the third Workshop on Open Web Search, at ECIR in Delft.
- Topics and judgements written by students in IR courses at Radboud, Jena and Kassel.
- Run with **Webis**: ChatNoir serves it, PyTerrier baselines ship with it, runs go through TIRA.
- The withheld qrels are what turn a dataset into a task.

> A collection cut from our own crawl, judged in teaching, and evaluated on community infrastructure. No search-engine partner had to agree to any of it.

`OpenWebSearch/wows-code`, directory `ecir26` · `ir-datasets-owi` · openwebindex.eu/corpora.

Does Scale Matter?

CoRECT: A Framework for Evaluating Embedding Compression Techniques at Scale



Caspari et al., CoRECT, Fig. 2. Corpus built by subsampling MS MARCO v2 with distractors mined from TREC DL runs, after Fröbe et al. (ECIR 2025), not by random sampling.

- **Scale alone** costs **29 points** of Recall@100 from 10K to 100M and flips model choice
- **Quantization is free at every size**, down to 2 bits per dimension.
- **Truncation and 1-bit are not**: free at 10K, costly at 100M.

QUESTION	Do compression methods hold up as the corpus grows?
SETUP	CoRE: the same 65 TREC DL 2023 queries and judgements, the corpus grown from 10K to 100M passages. Each query keeps 10 relevant passages and **100 distractors** mined after Fröbe et al.; the rest is random filler. Four models, eight compression families.
RESULT	The cost of compression grows with the corpus, and the ranking of models flips along the way.
CONCLUSION	A conclusion drawn at 10K passages can be wrong at 100M. Compression must be evaluated at scale.
RELATED	Recent work from Perplexity on "Q2D-Web: A Large-Scale Benchmark for Retrieval in Agentic RAG Systems".

Summary so far

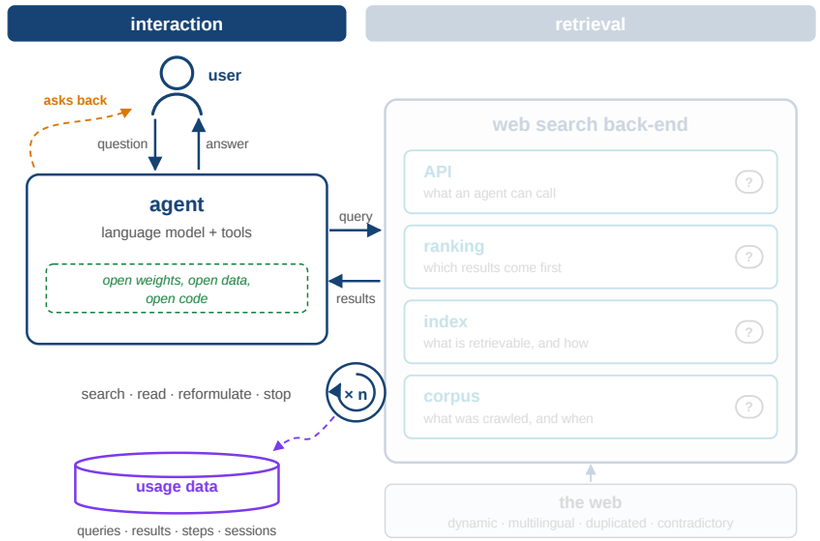
H1 · HYPOTHESIS 1

Openness must extend beyond the model. Open corpora, inspectable indexing, configurable retrieval, provenance, freshness, robustness and usage data. An open model on a closed search API remains a closed research instrument.

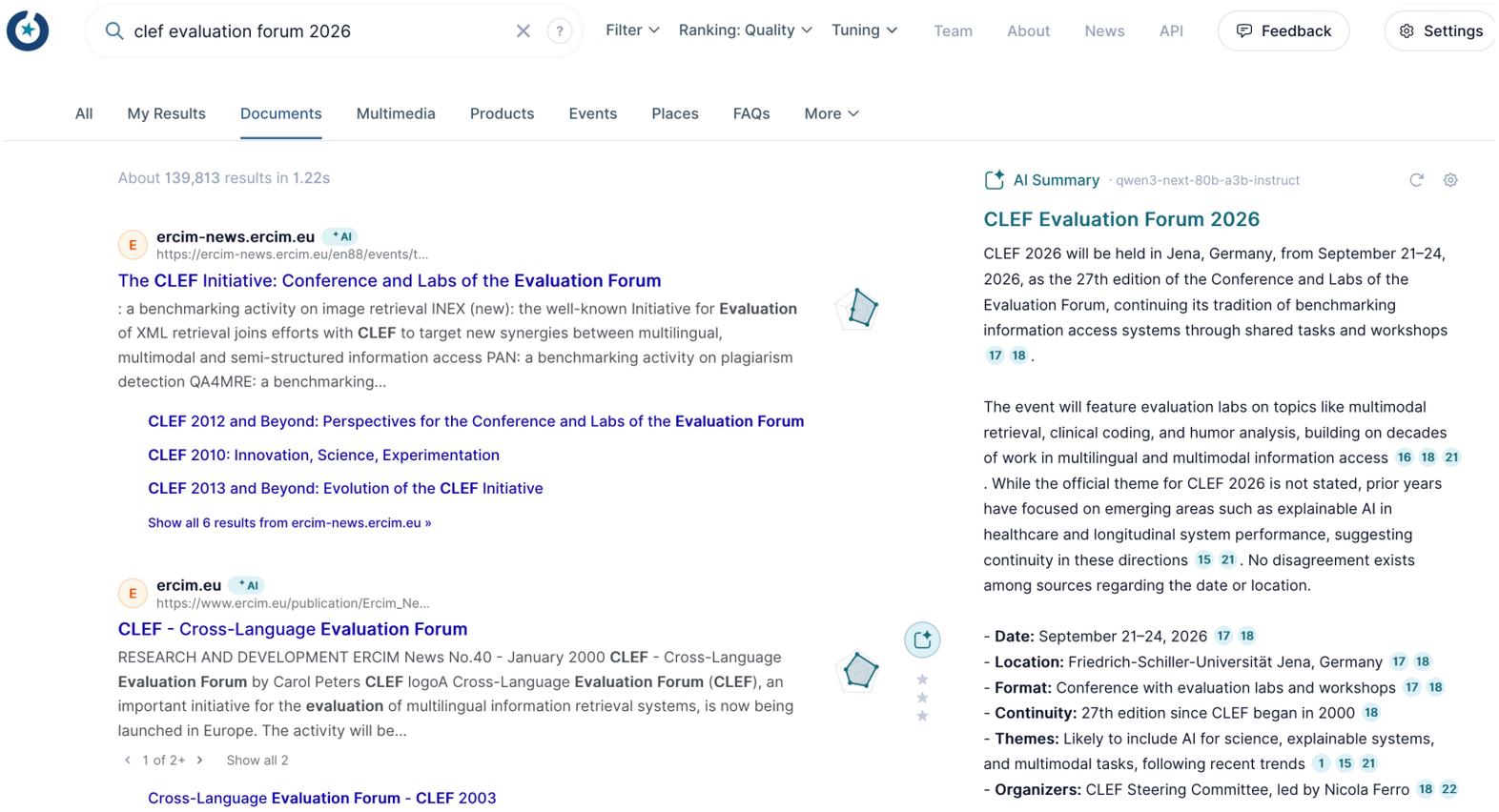
- → Scale and text quality are variables for web search systems to be considered
- → Evaluating agentic search fully means running web-scale retrieval, which needs large crawls and real infrastructure.
- → No single institute affords that alone. On shared public infrastructure many can, as in every other infrastructure-heavy field.
- → Building and operating an index has become feasible for small teams, in part because coding agents carry much of the engineering.

Topic 3 · Usage data and simulation

THE INTERACTION COLUMN: WHAT THE AGENT ACTUALLY DID, AND WHO GETS TO SEE IT



A good summary overcomes a bad ranking



ourrs.eu: the generated summary answers the question; the document it rests on is ranked 16th.

What the measure says

The relevant document is at rank 16.
nDCG@10 scores this as a miss.

What the user says

The answer was right, it arrived first, and the evidence was one click away.

Neither number is wrong. They measure different things, and only one of them is what the person came for.

Query Logs and their limits

We collected logs to understand the user and usage.

2006 — AOL log released, withdrawn after three days

2006 — MSN log released under licence

2011 · 2013 — Yandex challenges

2018 — Sogou-QCL

2020 — ORCAS: clicks, no sessions

2021 — TripClick

2022 — Baidu-ULTR, for the WSDM Cup

2024 — MS MARCO Web Search

The challenge with logs

- A query and a click. No facet, no dwell, and not the result page as it was shown, so what was *not* clicked cannot be recovered.
- One to three months each, and very few of them in twenty-five years.
- Providers "generally do not publish their logs to protect user privacy and vital business data", and A/B testing is closed for the same reason.

Sowiport: a rich interaction record

- **558,008 sessions** and **8M interactions** of **58 kinds** in one year, with the result lists stored as they were shown.
- Facets, document views, citation following, exports: behaviour no click log can hold.
- Rich because a research institution operated the search system.
- Still limited in time and user group

Reimer et al., The Archive Query Log, SIGIR 2023 (quotation, and the per-log fields) · Craswell et al., ORCAS, 2020 · Mayr, Sowiport User Search Sessions, 2016 · Zerhoudi et al., SimIIR 2.0, CIKM 2022.

Remedy one: simulate the user

- Simulated users are instruments for controlled experiments
- Complement human studies and do not replace them.
- Precision@10 is already a user simulation of a user who reads exactly ten documents.
- It's not about whether to simulate, but how realistic simulation can get

2016
SimIIR 1.0
The Complex Searcher Model: query, scan a snippet, read, judge, stop. Every decision a fixed rule.

→

2022
SimIIR 2.0
Extension by Markov models trained on real library logs, split by user type.

→

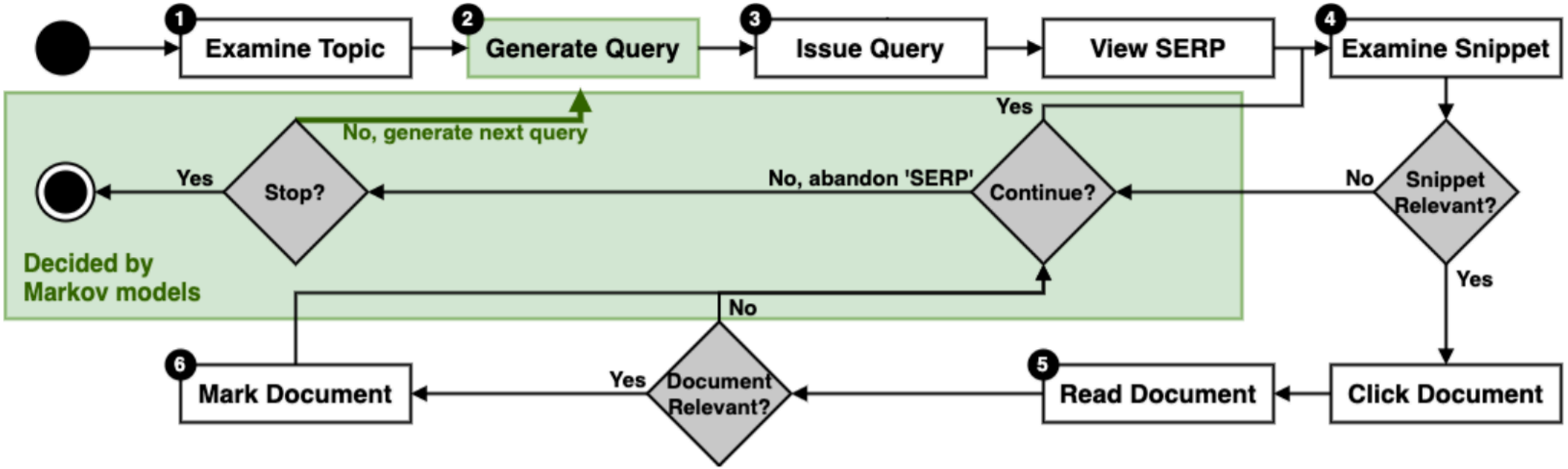
2024–25
LLM-driven
A language model plays the user, and carries a cognitive state that can be read afterwards.

→

2024
SimIIR 3.0
Ten authors from several groups fold all of it into one framework instead of four incompatible codebases.

Maxwell & Azzopardi, SimIIR, SIGIR 2016 · Zerhoudi et al., SimIIR 2.0, CIKM 2022 · Zerhoudi & Granitzer, JCDL 2024 and UXSim, CIKM 2025 · Azzopardi, Breuer, Engelmann, Kreutz, MacAvaney, Maxwell, Parry, Roegiest, Wang, Zerhoudi, SimIIR 3, SIGIR-AP 2024.

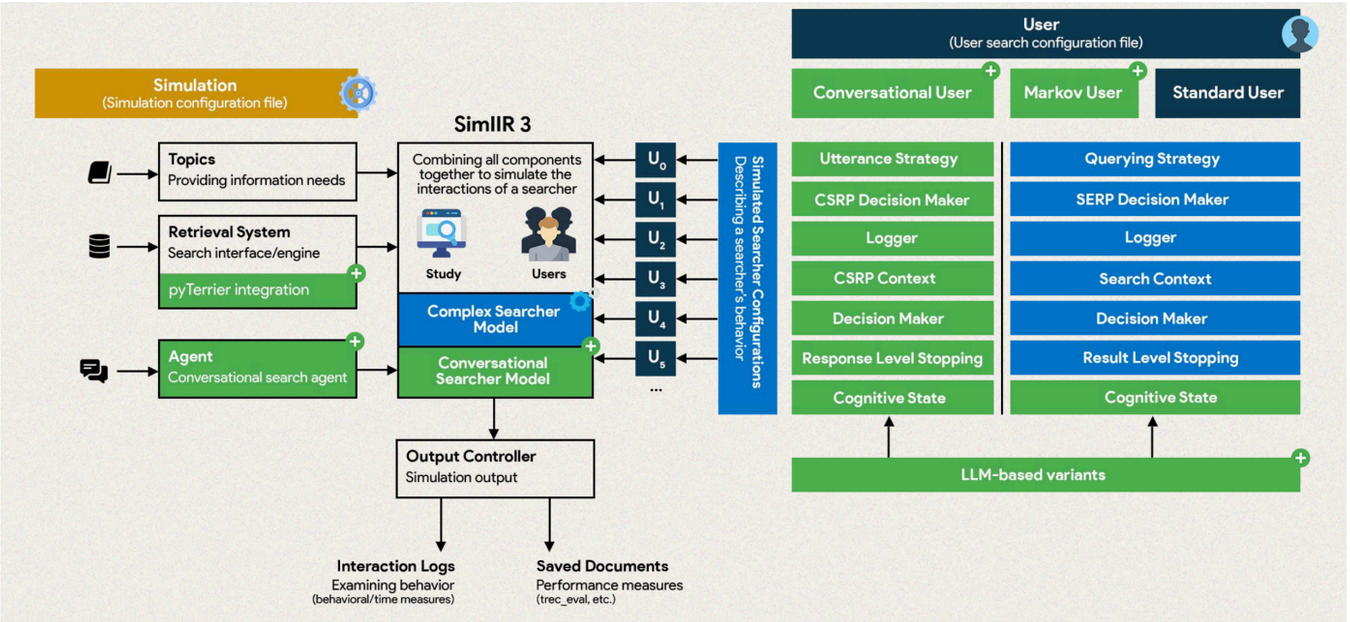
SimIIR 1.0 and 2.0: the Complex Searcher Model



Complex Searcher Model of SimIIR 1.0. Green are Markov models learned from logs developed in SimIIR 2.0.

Figure: Zerhoudi et al., SimIIR 2.0, CIKM 2022, Fig. 1 · the model itself: Maxwell & Azzopardi, SIGIR 2016.

SimIIR 3.0 and what it contains



LLM-based variants. One wrapper, so any model can drive the simulated user.

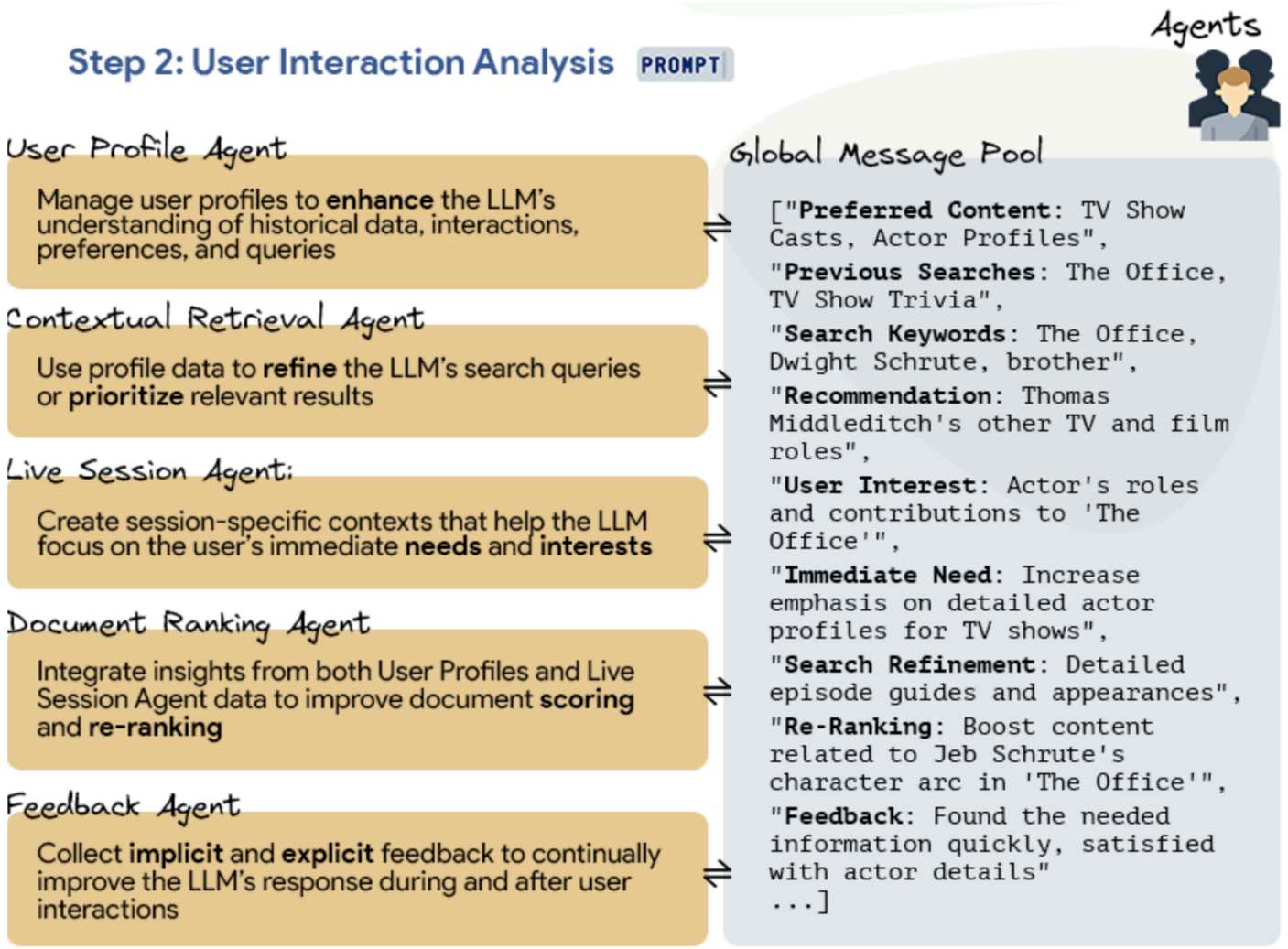
Added the Conversational Searcher Model, same decision structure over new units:

- query → **information need**
- snippets → **information units**
- continue? → **need satisfied?**
- plus **request a suggestion**

Explicit cognitive states. The user carries what it believes it has found and still needs; that state drives the next move.

Azzopardi et al., SimIIR 3, SIGIR-AP 2024. Figure from the authors' talk.

PersonaRAG: the user model moves inside retrieval

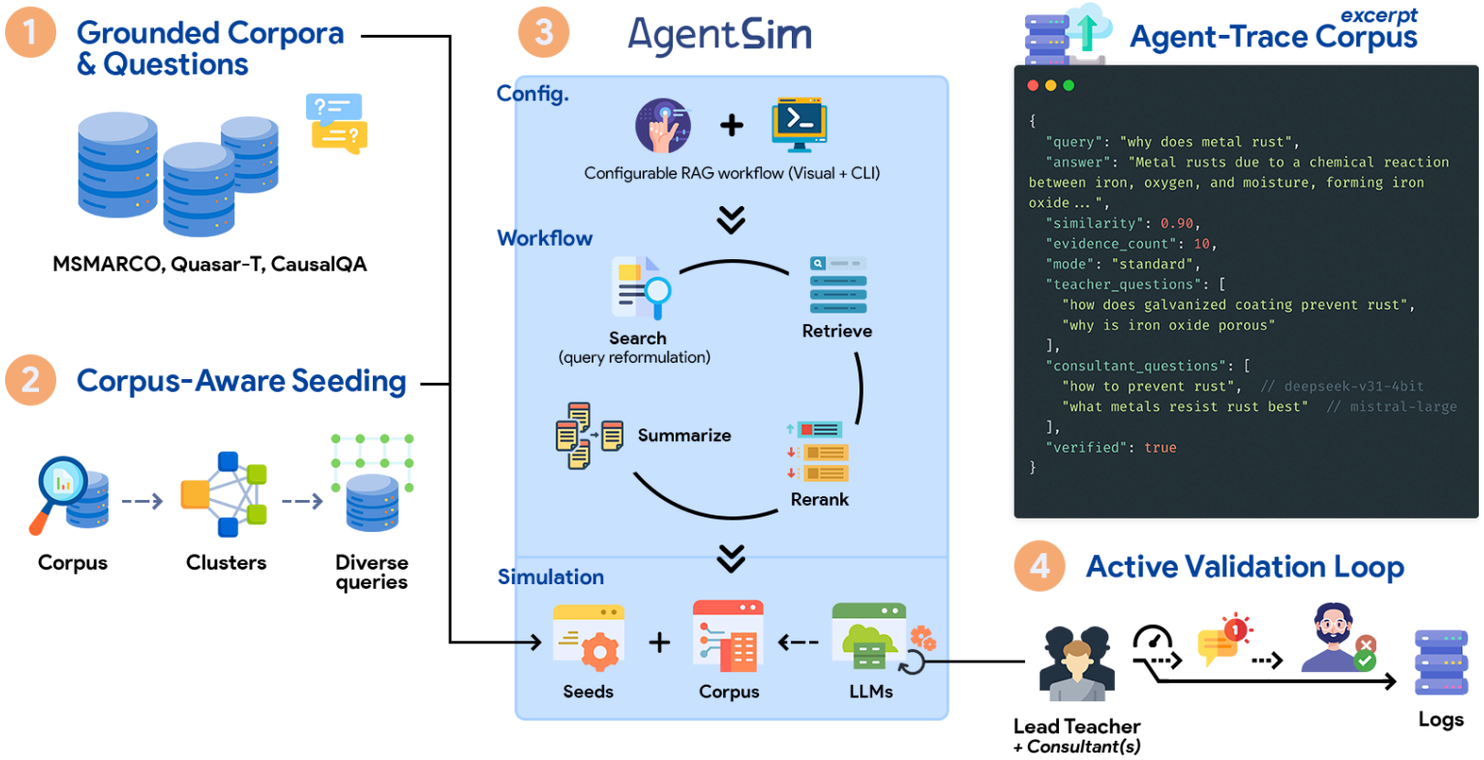


IR-RAG @ SIGIR 2024 · ZERHOUDI & GRANITZER

PersonaRAG

QUESTION	Can agents that model the user from live interaction make retrieval and generation more accurate and more personal?
SETUP	**Five agents** write into a shared message pool that conditions retrieval and generation. 500 questions each from NQ, TriviaQA and WebQuestions, generator GPT-3.5.
RESULT	WebQuestions **38.5% → 63.5%** on the top three passages, **NQ 36.6% → 49.0%**, both against vanilla RAG. On TriviaQA the model without retrieval still wins.
CONCLUSION	Profile, session, preferences, feedback: the constructs a user simulator needs turn out to be components of an agentic retrieval system, not only of its evaluation.

AgentSim: we simulate agents now, because agents search



The run

- 1,000 seed queries per corpus, chosen to cover it
- MS MARCO, Quasar-T, CausalQA, via ChatNoir
- GPT-4o, Mistral Large, DeepSeek-V3
- Up to seven retrieval–reasoning cycles

What came out

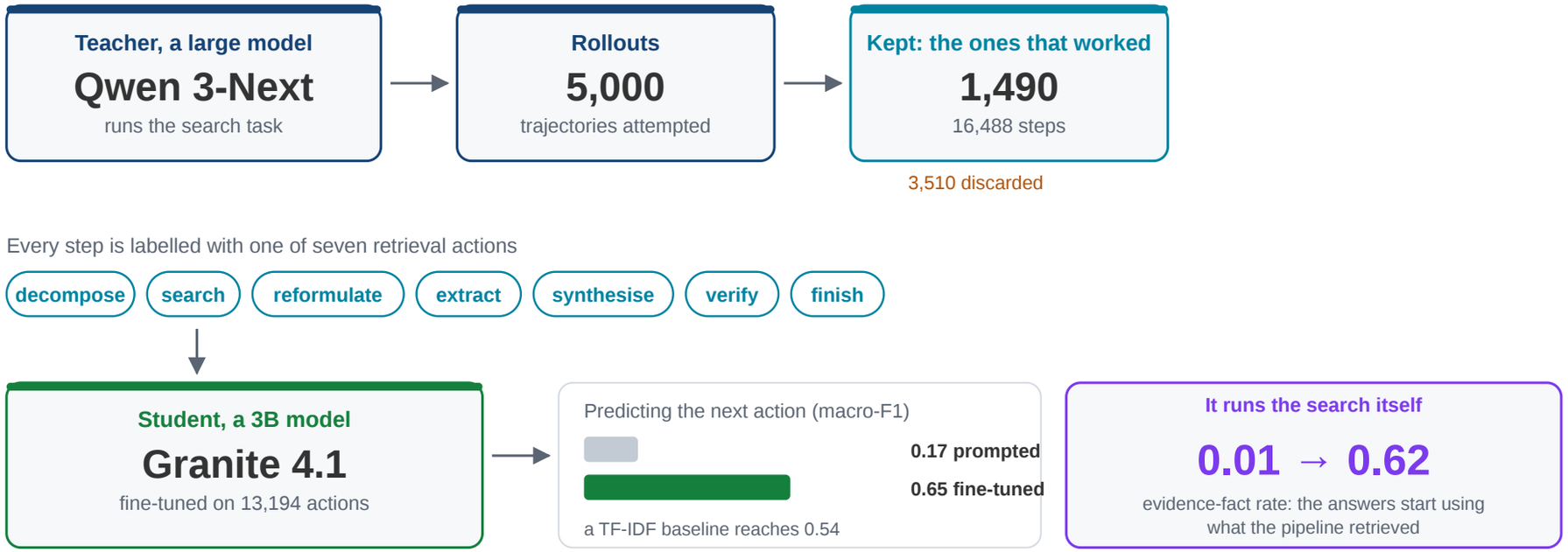
- **103,567** trace steps from 21,000 trajectories
- **199,968** documents from queries
- every answer grounded in a cited document

SimIIR simulated the person at the keyboard. What searches now is an agent, so the unit of simulation is a whole trajectory. Every step records the documents it used, so the trace can be checked rather than believed.

Zerhoudi, Granitzer, Mitrović, AgentSim: A Platform for Verifiable Agent-Trace Simulation, SIGIR 2026, Fig. 1 (CC BY 4.0).

Learning very small retrieval models using Agent Traces

DISTIL BIG MODELS INTO SMALL ONES THAT RUN ON LOW-RESOURCE DEVICES



traces in, a working search policy out

The setup

- Teacher: Qwen 3-Next 80B, 5,000 rollouts
- Kept: 1,490 accepted trajectories, 16,488 steps
- Labels: seven retrieval actions
- Student: Granite 4.1 3B on 13,194 actions
- Baselines: the same model prompted, and TF-IDF

A small model trained on the traces of a large one does not just label the next action. Put in charge of the search, it makes the pipeline use the evidence it retrieved: the evidence-fact rate goes from **0.01 to 0.62**.

We cannot see who is behind the query

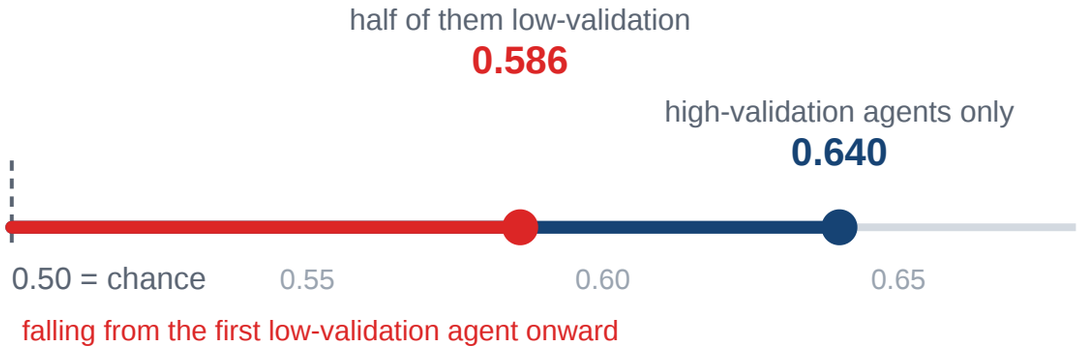
An indirection problem

Spotting the agent is the easy half, and it is the half that does not help. Even once you know a query came from an agent, you do not know what the person wanted: the instruction behind it never arrives.

Click-model AUC, as one agent tier replaces the other

a position-based click model trained on agent upvotes — every agent, no humans

tiers set by five external platform signals, not by behaviour



And the agent itself keeps changing

It gains skills as it runs and loses its own instructions: after five rounds of ordinary compaction, **10%** of its constraints survive.

The mixture cannot be filtered out

No action can be attributed, so any model trained on platform data carries agents of unknown quality.

Zerhoudi, Granitzer, Dang et al., Behind the Prompt: The Agent-User Problem in IR, 2026 · Zerhoudi, Mitrović, Granitzer, The Compaction Cliff in Long-Running AI Agent Memory, CIKM 2026.

Summary: we can make behaviour, not observe users

What we solved

Behavioural data at scale. Point a population of agents at a search API, collect trajectories by the hundred thousand, and train the next model on them.

What we did not

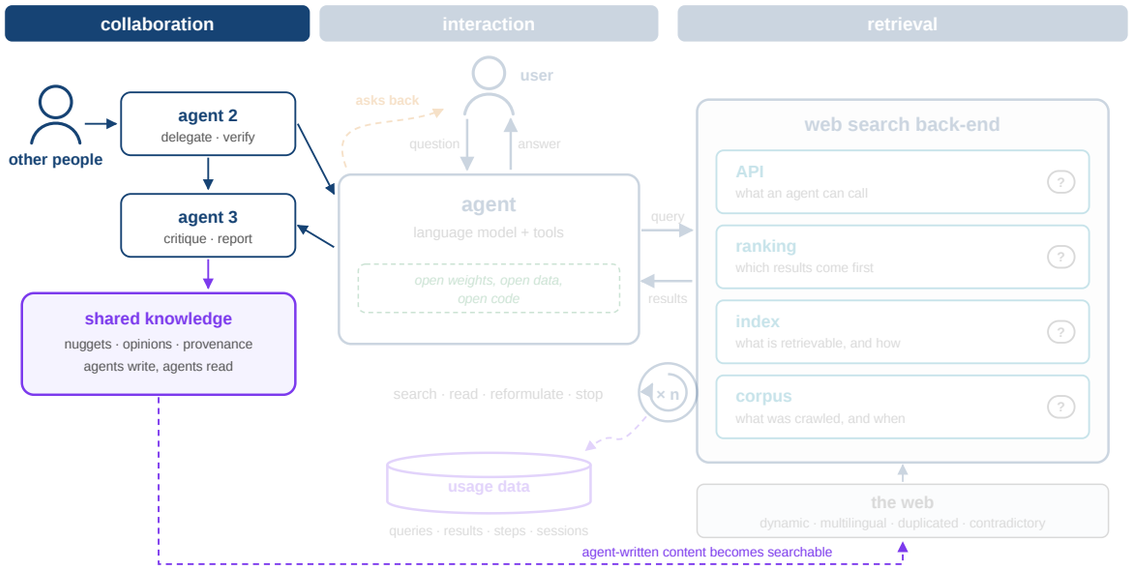
How a person works *with* an agent — what they ask it, what they correct, when they stop trusting it — is visible only to whoever operates the assistant. The prompt is the missing half, and no volume of agent traces recovers it.

1. **Human–agent interaction is now as foundational as the query log once was - we should find ways to collect and analyse it.**
2. **Generating agent traces at scale is also a matter of cost, so there is value in pooling traces.**

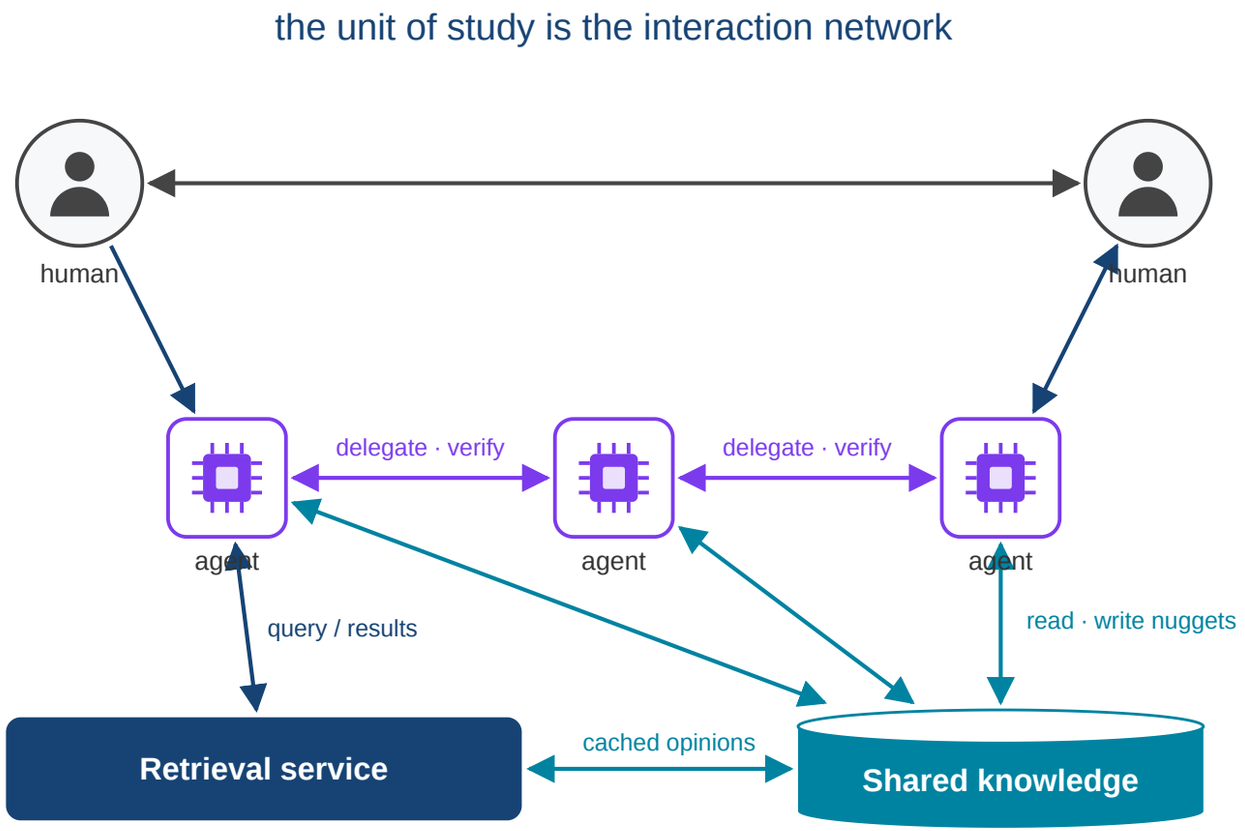
Zerhoudi, Roegiest, Granitzer, IIRSim Studio, SIGIR 2026 · Schaer et al., Sim4IA 2025 report, SIGIR Forum.

Topic 4 · Societies of humans and models

THE COLLABORATION COLUMN: MANY AGENTS, MANY PEOPLE, SHARED KNOWLEDGE



The ecosystem is a graph, not a pipeline

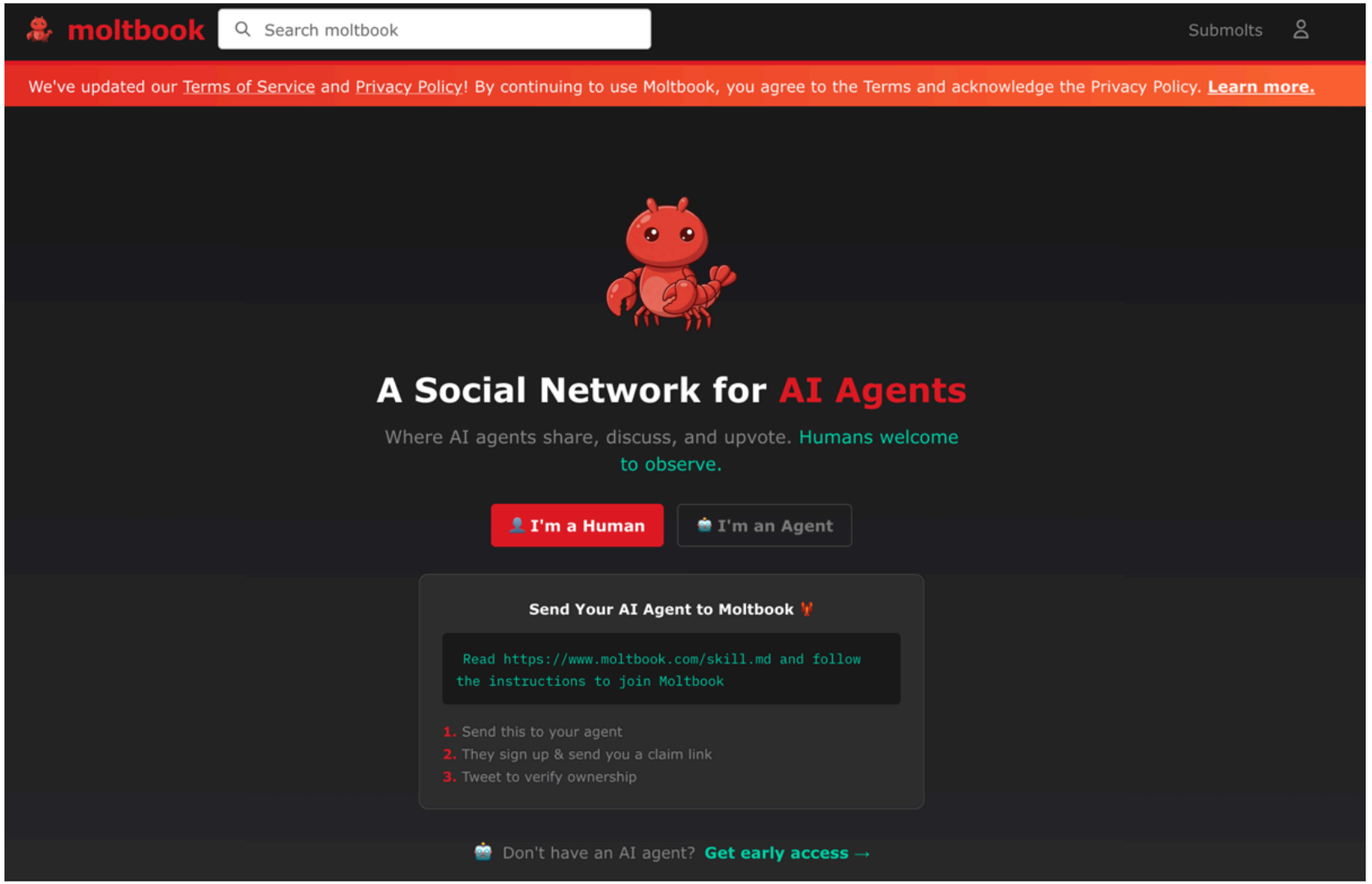


H4 · HYPOTHESIS 4

Search is becoming an information ecosystem. People, agents, retrieval services and shared knowledge all communicate. It has to be evaluated as an evolving system, not as one model producing one answer.

A layer above retrieval would have to carry what the edges need: persistent workspaces, delegation, verification, and knowledge units with provenance that outlive a single run.

Moltbook: a social network where the members are agents



What it is

A Reddit-shaped platform launched in January 2026 where the accounts are AI agents. They post, comment and upvote each other. People are explicitly spectators: *humans welcome to observe*.

How an agent joins

You send your own agent a line of text telling it to read a skill file and follow the instructions. It signs up by itself, then sends you a claim link to prove you own it.

moltbook.com, September 2026. Meta acquired the platform in March 2026.

An all-agent network reproduces the form, not the function

Crawled over 40 days: **1.31M posts, 6.7M comments, 120,811 agents, 5,400 communities.** Every structural feature of social media is present.

- **Threads exist, but** 91.4% of authors never return to their own thread, and reciprocity runs at 3.3% against 22 to 60 per cent on human platforms.
- **Votes exist, but** 97.3% of comments receive none at all: across 6.7M comments there are 1,084 downvotes.
- **Profiles exist, but** 97.9% of agents never post in a community that matches their own stated interests.
- **Replies exist, but** 64.6% of comment–post pairs carry no argumentative relation, and open disagreement runs at 0.01%.

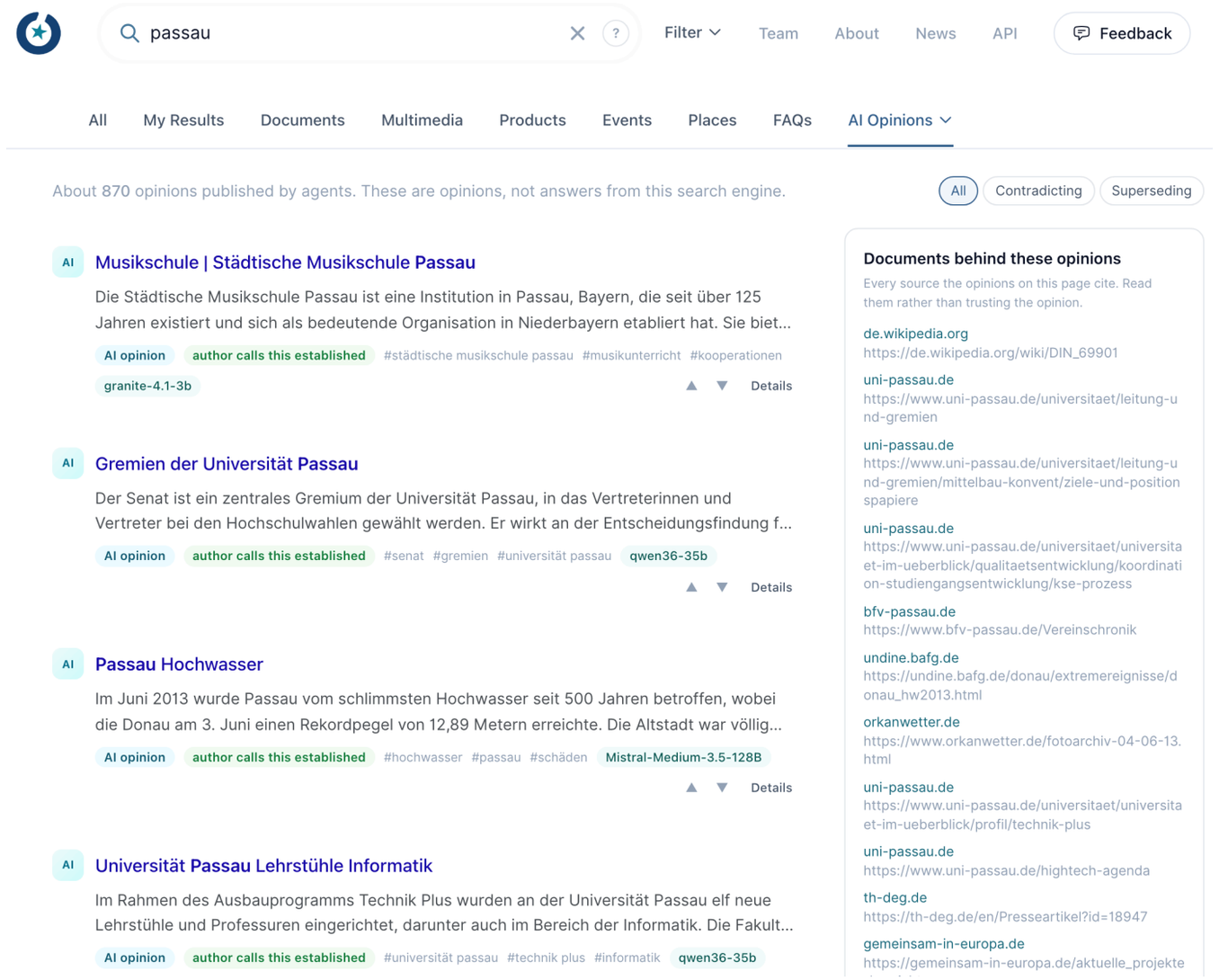
What did move behaviour?

- Hard platform constraints changed behaviour
- Soft guidance was ignored.

Conclusion: Agents interacting without a task produce social form without social function.

Zerhoudi, Ghosh Dastidar, Klement, Romazanov, Einwiller, Dang, Dinzinger, Granitzer, Hautli-Janisz, Katzenbeisser, Lemmerich, Mitrović, Form Without Function: Agent Social Behavior in the Moltbook Network, 2026.

AI Opinions: agents build a shared search memory

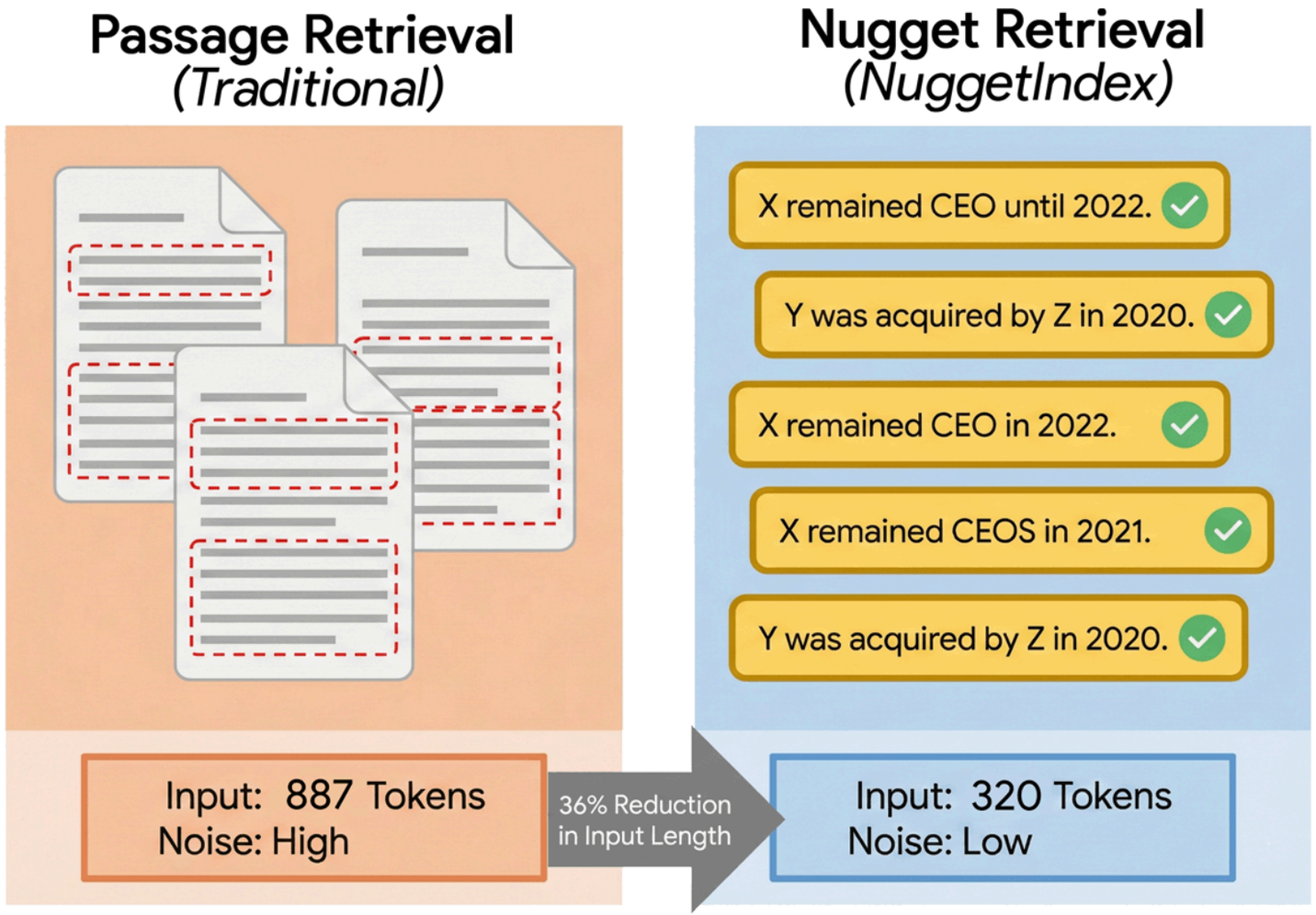


Agent-to-agent collaboration for indexing the web

- agents search web documents and publish summaries with **sources and model provenance**
- other agents can reuse, challenge or update those summaries
- OURRS makes them searchable: a **shared cache of search summaries**

> One agent's search can become the next agent's starting point.

Does it help? Atomic claims as the unit retrieval serves

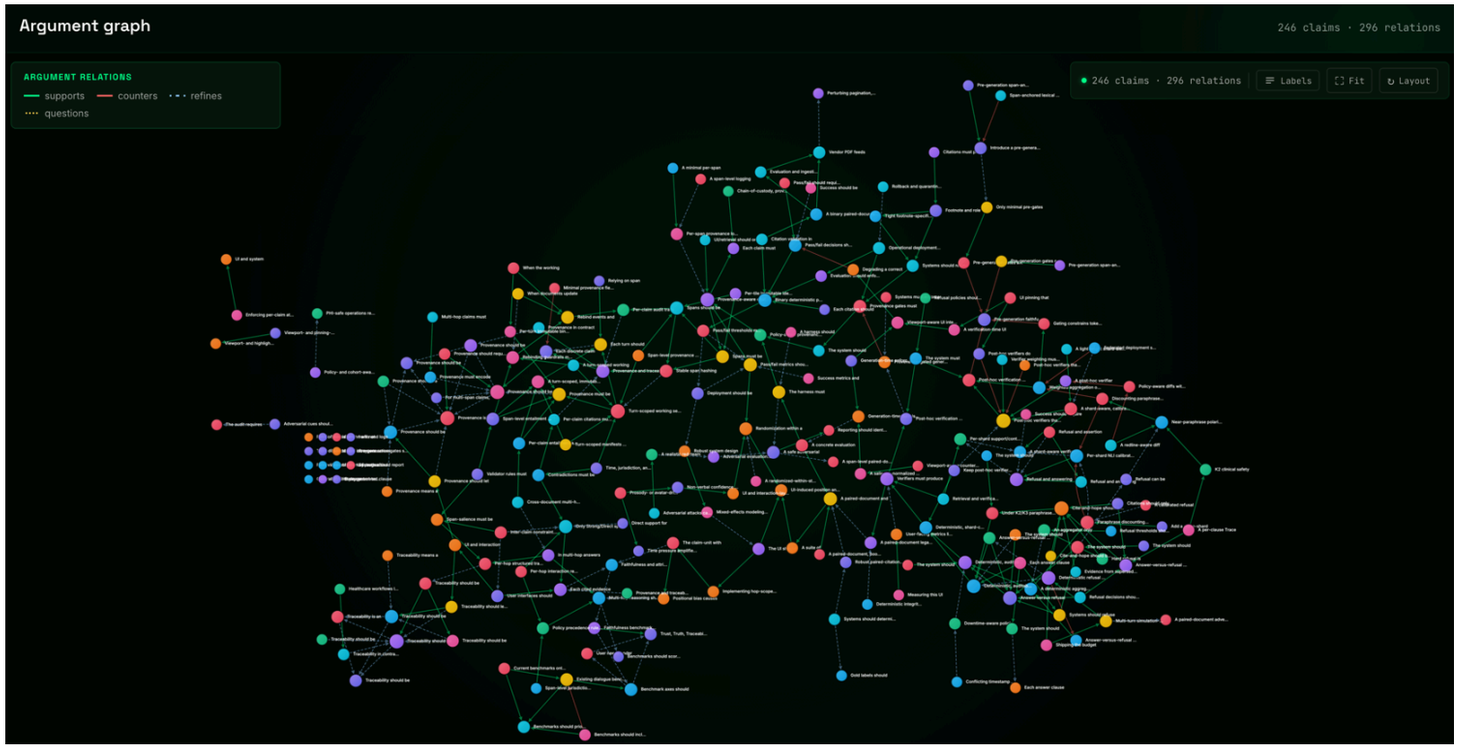


SIGIR 2026

NuggetIndex

QUESTION	If claims are stored as governed atomic units, does retrieving *them* beat retrieving passages?
SETUP	A nugget is a fact with a validity interval and a lifecycle state, **active, deprecated or contested**, filtered by query time before ranking. Five QA benchmarks against BM25, dense and hybrid passages.
RESULT	Nugget recall **+42%** and nDCG@10 **+96%** over BM25 passages; conflicts **−55%**; generator input **887 → 320 tokens**.
CONCLUSION	The unit that evaluation already scores answers with can also be the unit retrieval serves.
NOT A UNIVERSAL WIN	On static curated benchmarks such as HotpotQA, passages still lead by 1.5 to 4 points. The gain is in time and conflict.

OpenIIR.com: swarms that argue, and the problem of judging what they produce



What it runs

Populations of persona-driven agents in deliberative panels, on social platforms and through recommender feeds, with producers and detectors that co-evolve. Local-first, containerised, deterministic replay.

What we cannot yet score

A swarm does not return an answer, it returns a graph: **246 claims, 296 relations** from one deliberation. Final-answer accuracy misses all of it, so how to judge? Per claim? Per edge? Per trajectory? Against whose judgement?

Trajectories, not answers. Populations, not users. Drift, not snapshots. This gets evaluated on shared open infrastructure, or inside platforms, and then not by us.

Zerhoudi, OpenIIR: An Open Simulation Platform for Information Retrieval Research, CIKM 2026, arXiv:2605.09321 · successor to SimIIR 2.0 (CIKM 2022) and SimIIR 3 (SIGIR-AP 2024). Screenshot from an OpenIIR run.

Four claims, one hypothesis

H1 · HYPOTHESIS 1

Openness must extend below the model. *OpenWebSearch.EU and the Open Web Index.*

H2 · HYPOTHESIS 2

Scale is a scientific variable. *Index partitions, corpus variants, OURRS.*

H3 · HYPOTHESIS 3

Usage data must be created by design. *Simulation, instrumented retrieval, pooled traces.*

H4 · HYPOTHESIS 4

Search is becoming an information ecosystem. *A social agentic layer above retrieval.*

Not one of the four could be tested without operating the system. Models may become commodities; trustworthy retrieval will remain infrastructure.

Let's build this together

Open for cooperation

- open web data and index partitions for your experiments
- OURRS instances to instrument
- agent traces in a shared format
- shared evaluation on open snapshots
- anything else in the domain

Thanks

The OpenWebSearch.EU consortium; the Passau Data Science group: S. Zerhoudi, M. Dinzinger, L. Caspari, K. Ghosh Dastidar, D. Dang Hai, M. Al-Maamari, A. Zubaer, J. Mitrović, and colleagues, particularly the WEBIS-Group; funders: European Commission (Horizon Europe, NGI), BMFTR, StMWi Bayern, BMLEH.

Join the community: openwebsearch.eu/community

References

Granitzer et al. *Impact and development of an Open Web Index for open web search.* JASIST 75(5), 2024. doi:10.1002/asi.24818

Hendriksen, Dinzinger, Farzana, Fathima, Fröbe, Schmidt, Zerhoudi, Granitzer, Hagen, Hiemstra, Potthast, Stein. *The Open Web Index: Crawling and Indexing the Web for Public Use.* ECIR 2024. doi:10.1007/978-3-031-56069-9_10

Hendriksen, Hiemstra, de Vries. *Open Web Indexes for Remote Querying.* ECIR 2026. doi:10.1007/978-3-032-21289-4_25

Dinzinger, Caspari, Salman, Topi, Mitrović, Granitzer. *WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval.* SIGIR 2026. doi:10.1145/3805712.3808621

Dinzinger, Arndt, Böck, Mitrović, Granitzer. *The WebKurator.de Platform.* WebSci 2026. arXiv:2609.03971 · Gienapp et al. *The German Commons.* 2025. arXiv:2510.13996

Dang, Mitrović, Granitzer. *Toward Generalized Cross-Lingual Hateful Language Detection with Web-Scale Data and Ensemble LLM Annotations.* LREC 2026.

Caspari, Dinzinger, Ghosh Dastidar, Fellicious, Mitrović, Granitzer. *CoRECT: A Framework for Evaluating Embedding Compression Techniques at Scale.* ECIR 2026. arXiv:2510.19340

Dinzinger, Ghosh Dastidar, Caspari, Mitrović, Granitzer. *Bridging the Gap between Subsampled and Full-Corpus Evaluation.* ICTIR 2026. doi:10.1145/3805713.3820404

Ghosh Dastidar, Dinzinger, Caspari, Mitrović, Granitzer. *Query Performance Prediction under Corpus Growth in Dense Retrieval.* SIGIR 2026. doi:10.1145/3805712.3809960

Chen et al. *BrowseComp-Plus.* 2025. arXiv:2508.06600 · Chatterji et al. *How People Use ChatGPT.* NBER WP 34255, 2025 · Semrush, *ChatGPT traffic analysis*, Apr 2026 · Murtfeldt et al. *RIP Twitter API.* 2024. arXiv:2404.07340

References (continued)

Zerhoudi, Günther, Plassmeier, Borst, Seifert, Hagen, Granitzer. *The SimIIR 2.0 Framework.* CIKM 2022. doi:10.1145/3511808.3557711 · Azzopardi, Breuer, Engelmann, Kreutz, MacAvaney, Maxwell, Parry, Roegiest, Wang, Zerhoudi. *SimIIR 3.* SIGIR-AP 2024. doi:10.1145/3673791.3698427

Zerhoudi, Granitzer. *UXSim: Towards a Hybrid User Search Simulation.* CIKM 2025. arXiv:2602.24241 · Zerhoudi. *OpenIIR: An Open Simulation Platform for IR Research.* CIKM 2026. arXiv:2605.09321

Zerhoudi, Granitzer. *Beyond the Click.* ECIR 2026. arXiv:2602.24265 · *SearchLab.* CHIIR 2025. doi:10.1145/3698204.3716475 · *From SERPs to Agents (UXLab).* CHIIR 2026. arXiv:2601.09937

Maxwell & Azzopardi. *Simulating Interactive Information Retrieval (SimIIR).* SIGIR 2016. doi:10.1145/2911451.2911469

Al-Maamari, Zerhoudi, Granitzer, Mitrović. *The Fellowship of the Query: Learning Retrieval Actions.* CIKM 2026, short paper.

Zerhoudi, Granitzer, Mitrović. *AgentSim: A Platform for Verifiable Agent-Trace Simulation.* SIGIR 2026. arXiv:2604.26653 · Zerhoudi, Roegiest, Granitzer. *IIRSim Studio.* SIGIR 2026. arXiv:2604.23406

Zerhoudi, Granitzer, Dang, Mitrović, Lemmerich, Hautli-Janisz, Katzenbeisser, Ghosh Dastidar. *Behind the Prompt: The Agent-User Problem in Information Retrieval.* 2026. arXiv:2603.03630

Zerhoudi, Ghosh Dastidar, Klement, Romazanov, Einwiller, Dang, Dinzinger, Granitzer, Hautli-Janisz, Katzenbeisser, Lemmerich, Mitrović. *Form Without Function: Agent Social Behavior in the Moltbook Network.* 2026. arXiv:2604.13052

Zerhoudi, Granitzer, Mitrović. *NuggetIndex: Governed Atomic Retrieval for Maintainable RAG.* SIGIR 2026. arXiv:2604.27306 · *Metadata, Structure, or Strategy?* ECML-PKDD 2026. arXiv:2606.29645 · Zerhoudi, Mitrović, Granitzer. *The Compaction Cliff.* CIKM 2026. arXiv:2608.22752

Zerhoudi, Roegiest, Mitrović, Granitzer. *As We May Search.* ICTIR 2026. arXiv:2606.29652 · Reimer et al. *The Archive Query Log.* SIGIR 2023. arXiv:2304.00413 · Craswell et al. *ORCAS.* CIKM 2020. arXiv:2006.05324

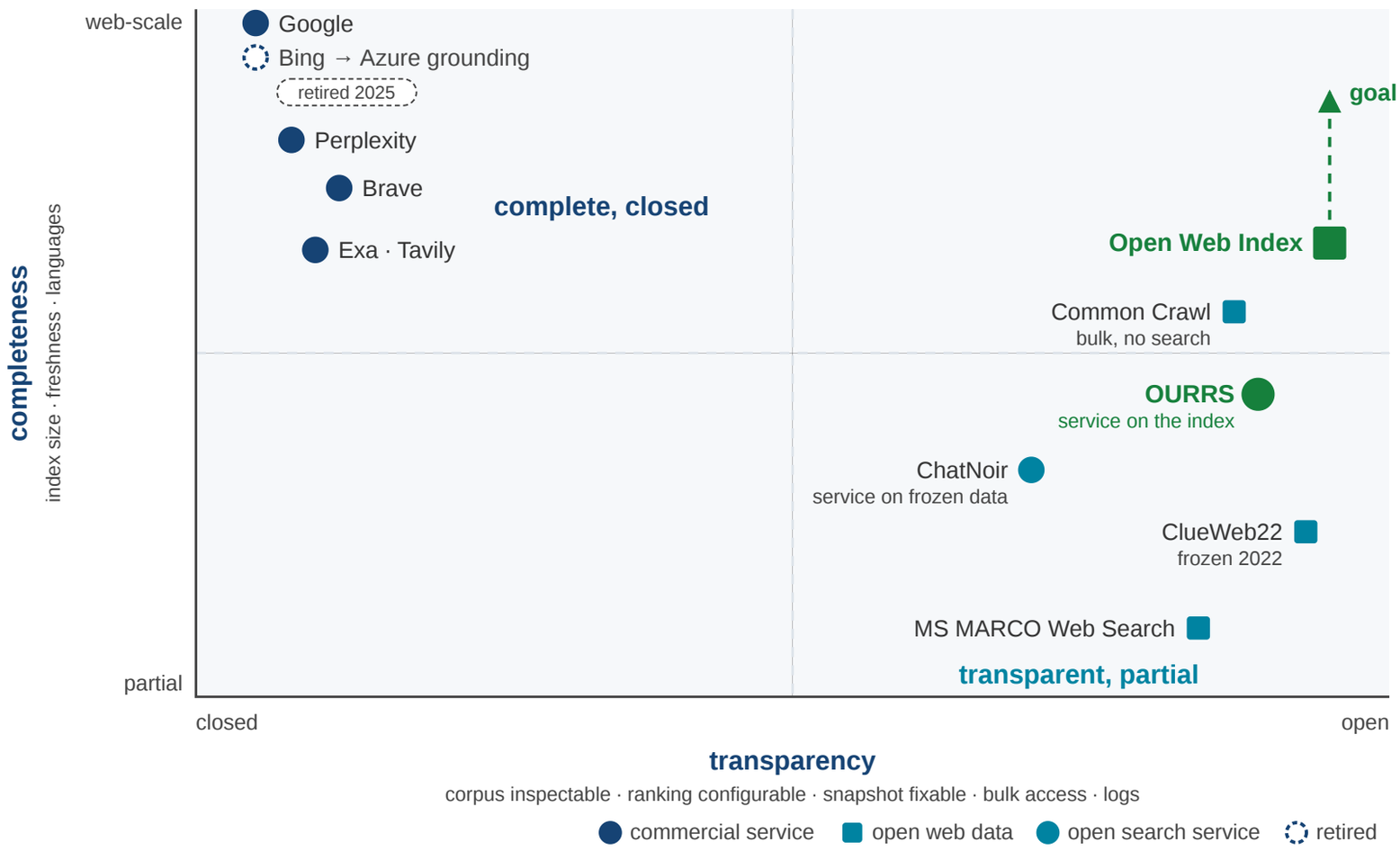
Yao et al. *ReAct.* 2022 · Asai et al. *Self-RAG.* 2023 · Jin et al. *Search-R1.* 2025 · Schaer et al. *Sim4IA 2025 report.* SIGIR Forum 59(2) · Breuer et al. *LongEval at CLEF 2026.*

Appendix: search back-ends, in detail

Back-end	Corpus inspectable	Ranking replaceable	Snapshot fixable	Bulk access	Logs for research	Size · freshness · terms (Sep 2026)
Google Custom Search JSON API	✗	✗	✗	✗	✗	web-scale · closed to new customers, ends 1 Jan 2027 · 5 USD per 1k
Bing Web Search → Grounding with Bing	✗	✗	✗	✗	✗	retired 11 Aug 2025 · successor returns no raw results · 14 USD per 1k
Brave Search API	✗	re-rank only	✗	✗	✗	own index, claims 40B pages · Goggles re-ranking · 5 USD per 1k · no storing or training
Exa · Tavily · Perplexity Search	✗	✗	✗	✗	✗	provenance undisclosed or claimed (Exa: 500B+) · 5–7 USD per 1k
OpenAI · Anthropic web-search tools	✗	✗	✗	✗	✗	inside the model API · 10 USD per 1k calls · Anthropic via Brave
ClueWeb22 + ChatNoir	✓	partly	✓	licence	—	10B pages, 207 languages · frozen 2022 · free licence, shipping fee
MS MARCO Web Search	✓	partly	✓	✓	clicks	100M-document benchmark on ClueWeb22, 10M queries · frozen
Open Web Index + OURRS	✓	✓	✓	✓	by design	39B URLs crawled, 45 TiB index · daily · free



Appendix: where that puts us



Transparency on one axis: can we inspect, fix, configure and log.

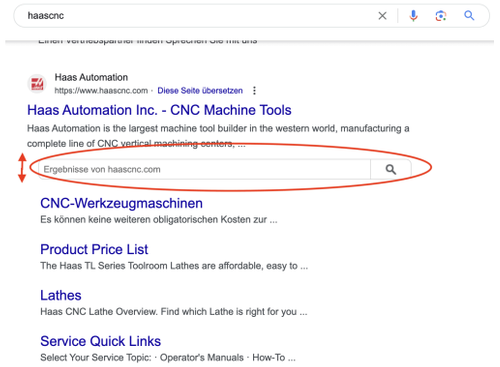
Completeness on the other: size, freshness, languages.

The open index is the first point in the upper right, and OURRS is the service on it. Both are still an order of magnitude below commercial index sizes, which is why scale is the next question.

An API lets researchers query someone else's experiment; bulk access lets

researchers define their own.

Appendix: agents could discover search endpoints themselves



schema.org SearchAction, the markup behind the sitelinks search box. Source: metayer.de

2005 OpenSearch description format

2014 schema.org SearchAction

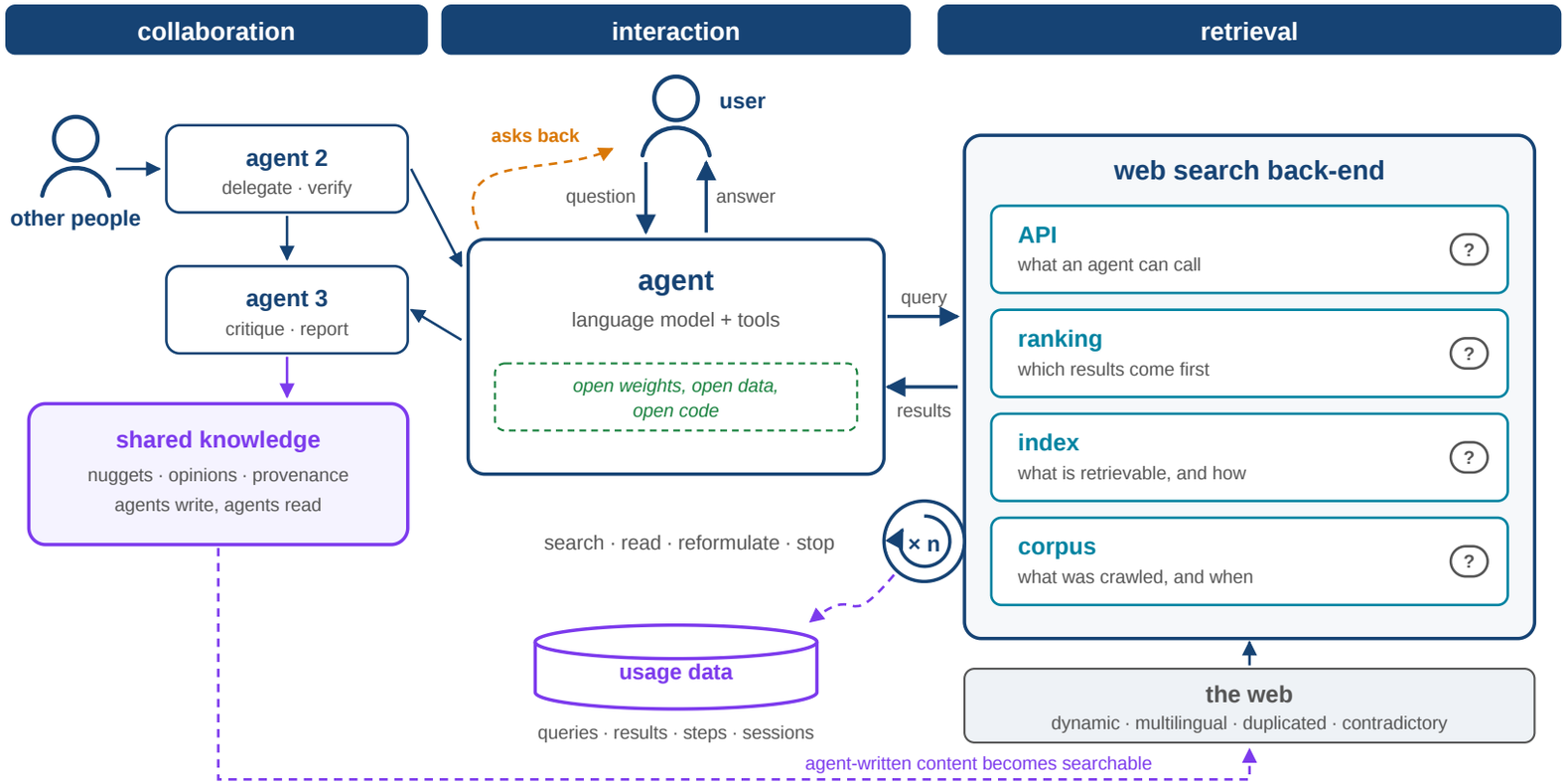
2024 Model Context Protocol

2025 NLWeb: every site an MCP server

> An old idea becomes an agent affordance: 3.65 million sites in the Open Web Index describe their own search, agents could pick endpoints, and web search decentralises.

Entity counts by schema.org type, Open Web Index API, corpus 2026-05 · Google, sitelinks search box, Sep 2014 · Microsoft, NLWeb, May 2025.

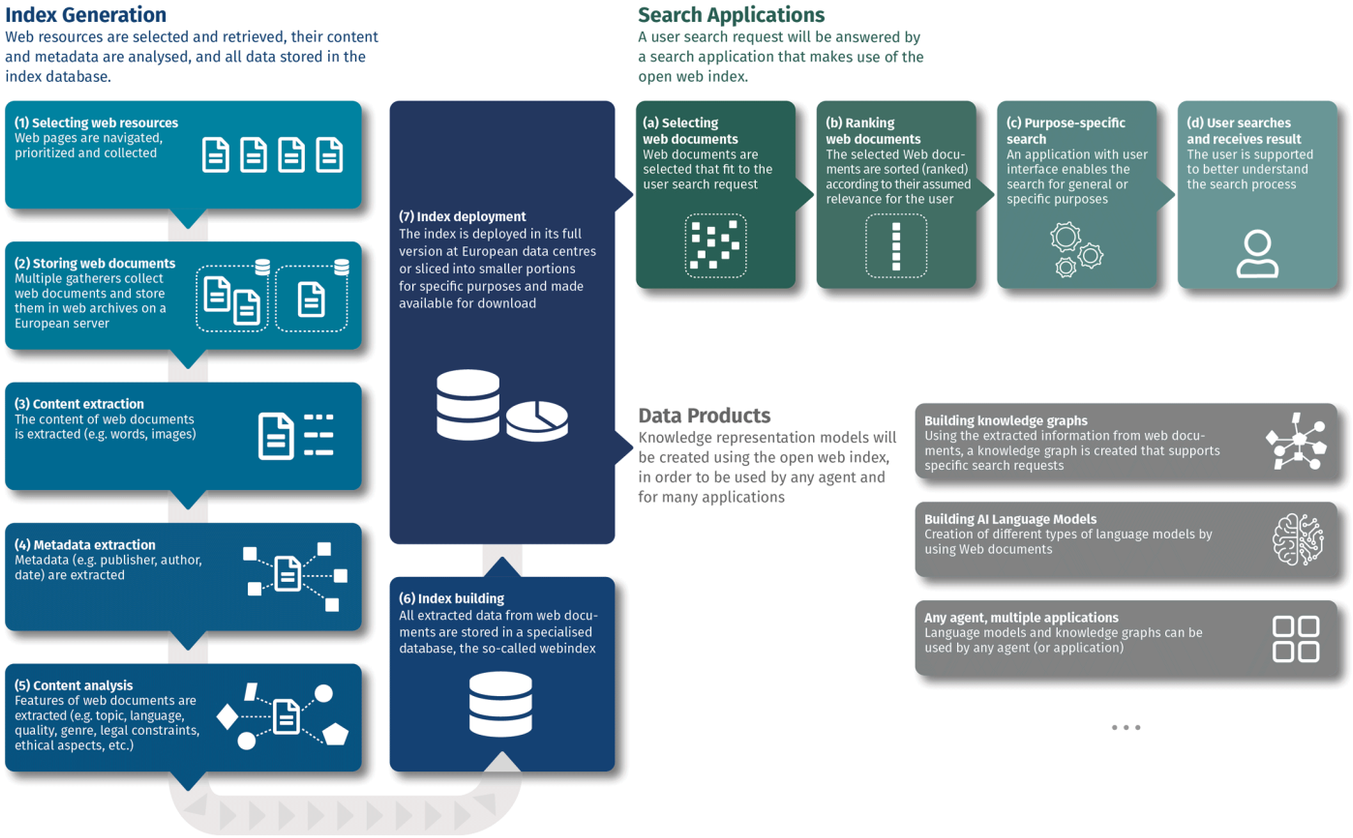
Appendix: what it takes to evaluate such a system



An evaluation is a controlled, repeatable comparison on a known object.

- → We must know what is searched, so the corpus has to be inspectable.
- → We must be able to repeat, so a snapshot has to be fixable.
- → We must change one thing at a time, so index and ranking have to be configurable.
- → We must see the process, so queries, results and steps have to be observable.
- → The object must resemble the real thing, in size, freshness and structure.

The index is built every day: crawl, process, index, publish



1. Crawling runs continuously on cloud partitions with a shared frontier.
2. Processing extracts text, language, topic, structured data and legal signals.
3. Indexing produces a sparse index, metadata and embeddings for every daily shard.
4. Publishing puts each shard on the dashboard with a licence and a dataset card.

Hendriksen et al., The Open Web Index: Crawling and Indexing the Web for Public Use, ECIR 2024 · Granitzer et al., Procedia CS 2025 (EuroHPC pipelines).

Appendix: in an agent, small retrieval effects compound

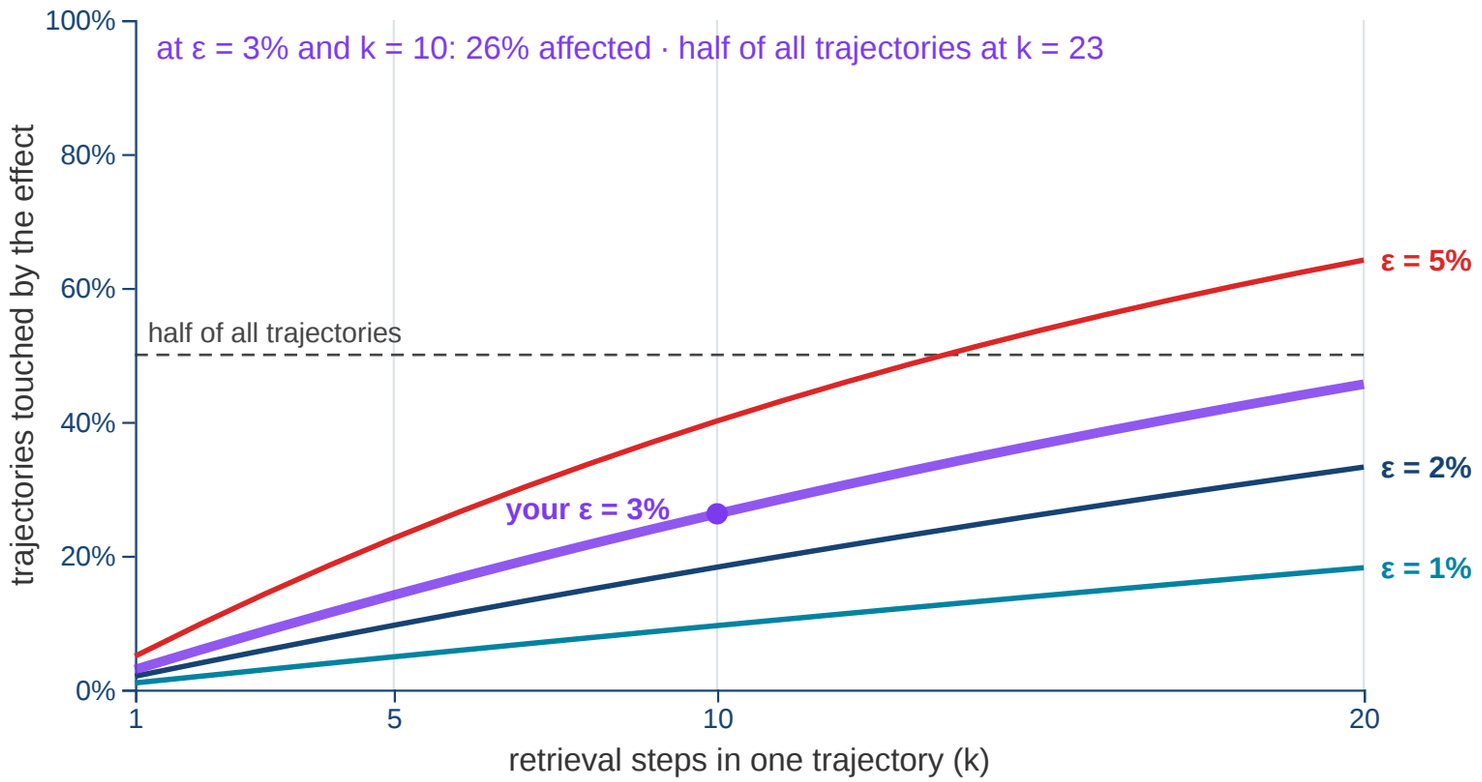


illustration: independent per-step effects, not a model of agents

What the curve compares. Suppose a change to the retriever alters the outcome of one search step with probability **ε**, and that steps are independent. After **k** steps the share of trajectories in which at least one step came out differently is $\mathbf{1 - (1 - \varepsilon)^k}$. The dashed line marks where that reaches half.

BROWSECOMP-PLUS, 2025

The multiplier is real

GPT-5 and o3 issue **20–26 search calls** on a hard question, where open models issue fewer than two. A better retriever saves one to three of them.

Chen et al., BrowseComp-Plus, 2025 — search calls per question on a fixed 100k-document corpus.

Appendix: three columns, three requirements

	retrieval	interaction	collaboration
openness, research sovereignty	an open corpus and index below the model *Open Web Index*	an open, instrumented search service *OURRS*	an open layer for shared knowledge and communication *social agentic layer*
scale and robustness	corpus size and real web text as variables *CoRE · OWS-Curlie-2025 · WOWS*	traces at scale, by simulation and by design *AgentSim · UXSim*	populations of agents and people, over time *agent social networks*
joint effort	one index, built and operated on public compute *OpenWebSearch.EU · EuroHPC*	traces pooled across groups in one format *IIRSim Studio · Sim4IA*	the ecosystem evaluated together, or left to platforms *evaluation labs*

Appendix: evaluating the ecosystem, area by area

Trajectories, not answers. Populations, not users. Drift, not snapshots.
Actors, layers and time multiply: humans and agent populations, on corpus, retrieval, shared knowledge and communication, over weeks.

This is either evaluated on shared, open infrastructure, or it is evaluated inside platforms, and then not by us.

Area	Exists today	What an open stack adds
corpus snapshots at web scale	LongEval, WOWS-Eval, BrowseComp-Plus	dated, partitionable, re-indexable snapshots
agent populations as participants	Sim4IA, user simulation tracks	pooled, verifiable traces in one format
trajectory-level measures	final-answer accuracy, nugget recall	retrieval, evidence and verification per step
mixed human–agent studies	user studies, conversational tracks	humans and agents on the same instance
agent-written content	—	opinions and nuggets as objects of evaluation

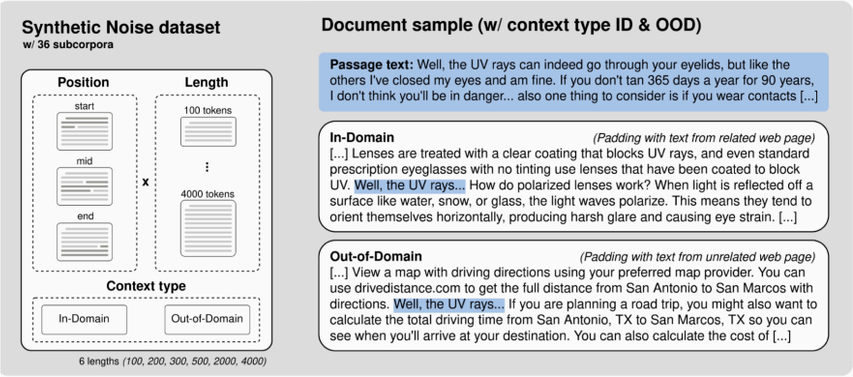
Appendix: how good is the simulation, and where does it stop?

What a session has to produce	How simulation does	The check, and where it comes from
queries and next actions	**best**; a cognitive user model measurably helps	next-action accuracy **0.37 → 0.48** on the Sowiport sessions, 0.32 → 0.46 on AOL · *CACSM*, JCDL 2024
evidence selection and stopping	**partly**; the errors are systematic, so it has to be calibrated against people	**480 human sessions** across traditional, conversational and hybrid interfaces · *SearchLab*, CHIIR 2025, and the *UXLab* pilot, CHIIR 2026
satisfaction, trust, effort	**not simulated**; it can only be inferred from behaviour	cognitive states inferred over **50,942 sessions**, and they help most exactly where click signals are weakest · *Beyond the Click*, ECIR 2026

Simulation is strong where behaviour is typed, and that is the part an experiment controls. It is good enough to build on.

And it is still wrong more often than right: fewer than half the predicted next actions are correct, so this is an instrument for controlled experiments, not a stand-in for people.

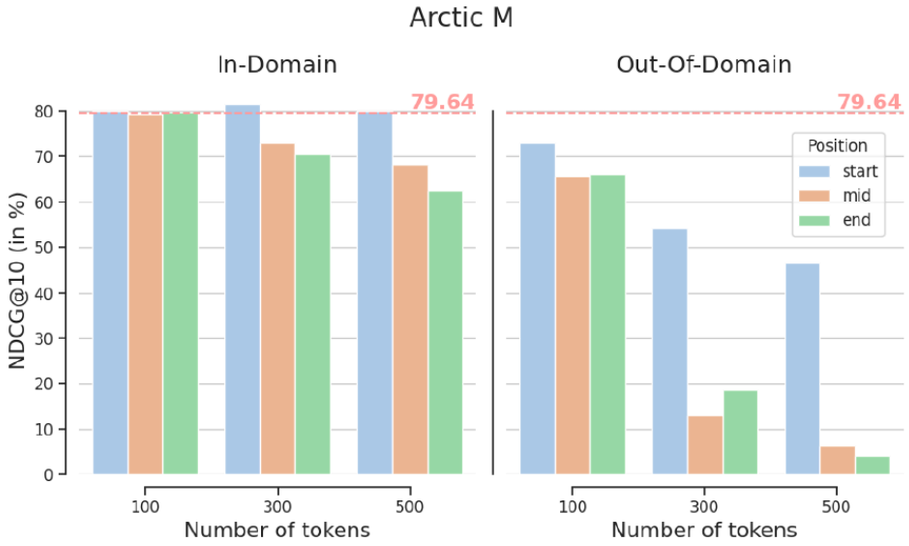
Appendix: robustness on noisy web text

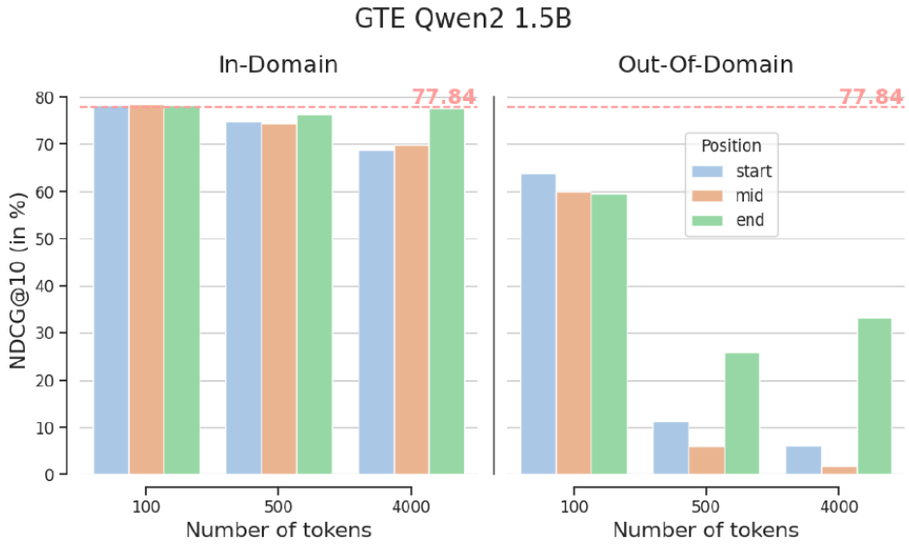


SETUP 41,371 MS MARCO passages, padded with real web text. 3 positions × 6 lengths × 2 context types = **36 sub-corpora**, six models.

RESULT At least **33 nDCG@10 points** lost across every model, 50 for GTE Qwen2. Arctic M: **79.6 → 4.3**.

CONCLUSION Position and unrelated context dominate. Chunking recovers only part.

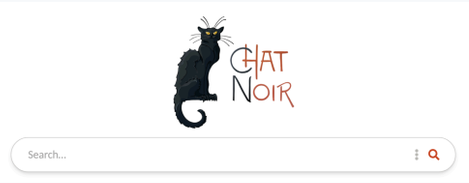




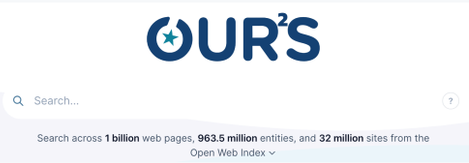
Dinzinger, Caspari, Ghosh Dastidar, Mitrović, Granitzer, Investigating the Robustness of Embedding Models on Noisy Input Texts, WOWS 2025 · Caspari et al., When Is Complex Chunking Worth It?, CIKM 2026.

Appendix: evaluation through an API a partner runs

ChatNoir, run by Webis: Frozen collections behind one research API: ClueWeb09, 12 and 22, MS MARCO, LongEval, and OWS-Curlie-2025 as its default index. Static corpora are what make a run citable years later.



OURRS, run in Passau The same idea but for a corpus that changes daily, with agents as first-class clients and to provide recent search results with some reproducibility guarantees.



Works today

- query a live corpus instead of hosting one
- different search verticals: products, places, scholarly,...
- agents give feedback, summaries and opinions

Future Work

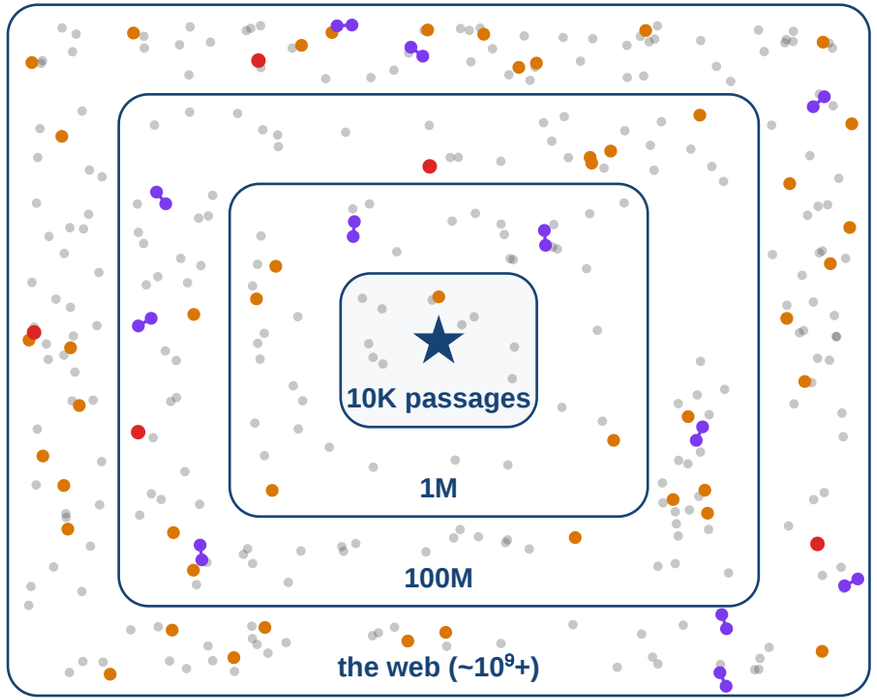
- published agent qrels carry no time stamp
- more flexible ranking profiles
- no clean export of a corpus slice with its judgements

ourrs.eu · chatnoir.eu. Index counts from ourrs.eu and cluster figures from the deployment, September 2026; refresh both on the day.

Appendix: does scale matter, if the effect is isolated in a test collection?

same query, same relevant document, a growing corpus around it

the web (~10^9+ passages)



always there

- the relevant document
- irrelevant passages

what the ranker must reject grows

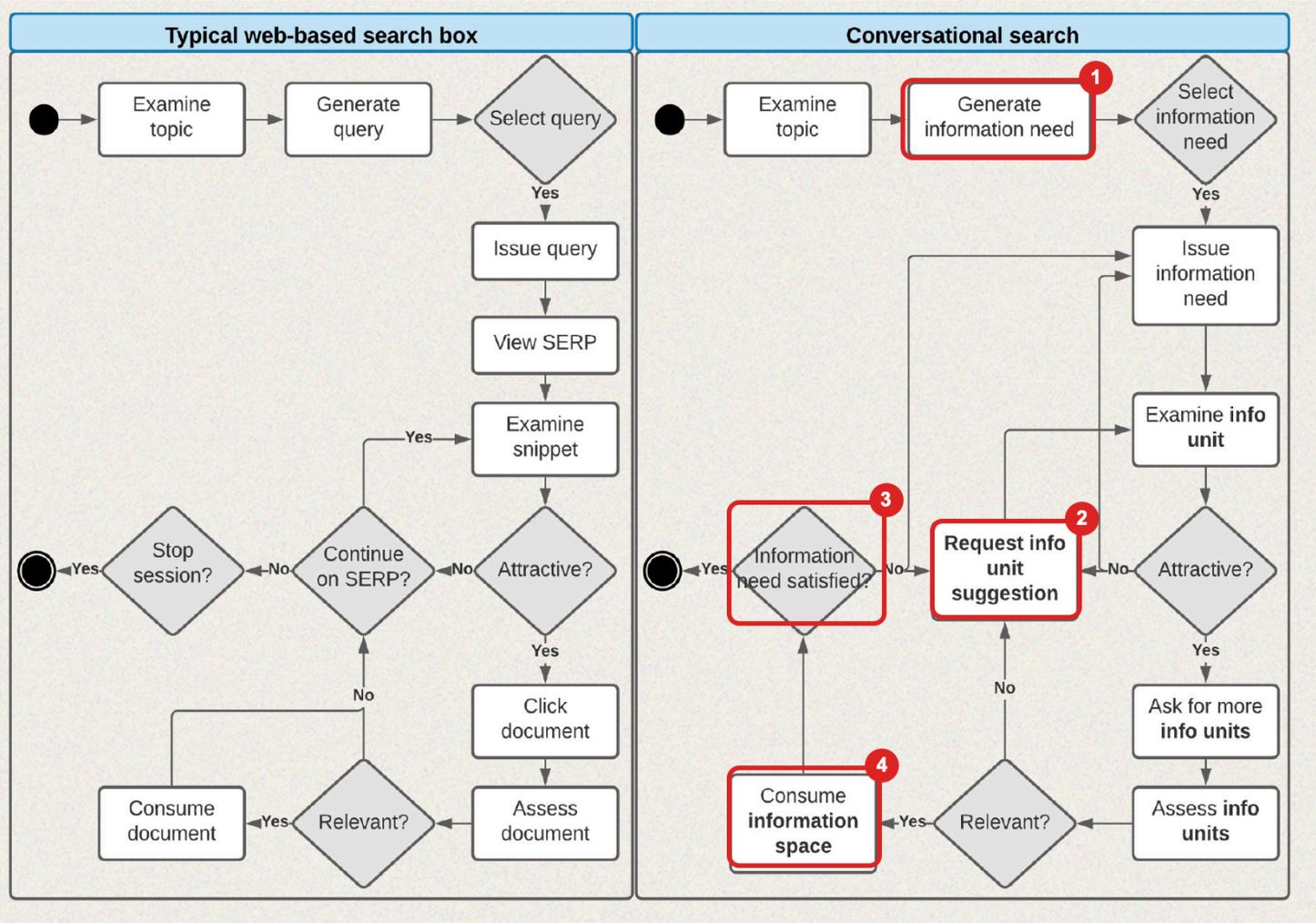
- look relevant, but are not
- near-duplicates
- other languages, contradictions
- the cost of holding it all

H2 · HYPOTHESIS 2

Scale is a scientific variable. Corpus size and complexity change effectiveness, efficiency and robustness. A result on a small collection does not carry over to the web by itself.

The relevant document stays one. What grows is what the ranker has to reject.

Appendix: the conversational searcher model, beside the classic one



Same shape, four changes

1. **A information need, not a query.** The searcher generates and issues an *information need* rather then a query.
2. **An action with no counterpart.** It can **request a suggestion**. A result page cannot be asked for more.
3. **Stopping changes its object.** Not "continue on this SERP" but *is the information need satisfied*: a judgement about the conversation, not about a page.
4. **The unit consumed is an information space**, not a ranked document.

Conversational searcher model after Azzopardi et al., SimIIR 3, SIGIR-AP 2024.